Legacy API (Banking & LAN) Host Commands

Cryptosec Dekaton
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Utimaco Iberia, S.L.U.

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Chapter 1

Local Master Keys

1.1 Introduction

The Local Master Keys (LMKs) are keys used in external storage, to encrypt operational data (keys, PIN blocks, decimalization tables) to be stored in a database.

There are two kinds of LMKs:

- TDES3: used in the Container Version 0 (or Variant) protection scheme.
- AES-256: used in the Container Version 1 (or Key Block) protection scheme.

1.2 TDES3 LMKs

The TDES3 Local Master Keys are triple length TDES keys derived from the TDES3 Master Key. The TDES3 Master Key is loaded in shares through the console interface.

Each TDES3 LMK key is associated to a given purpose. Keys stored in databases will be associated with one LMK Key. Thus, the keys are grouped by their purpose. Grouping the keys ensures that a key is not used for anything else than its intended purpose.

Table 1.1: TDES3 LMKs definition

LMK	Description	Operation	Authorization Mode Only
0	TDES3 Master Key	KCV calculated.	No
1	Deprecated	N/A	N/A

TDES3 LMKs definition (continued)

LMK	Description	Operation	Authorization Mode Only
2	Key Transport Keys	Transported in shares (when symmetric).	Yes
		Generated. Deleted (when symmetric). Wrapped (with LMK2 key). Encapsulated (when symmetric). Key wrapping (when symmetric). Key encapsulation (when asymmetric). KCV calculated.	No
3[1]	PIN Block Transport Keys	Transported in shares.	Yes
		Generated. Deleted. Wrapped (with LMK2 key). Encapsulated. Diversified. KCV calculated. PIN block transport. PIN block LMK encryption.[2]	No
4[1]	PIN Verification Keys	Transported in shares.	Yes
		Generated. Deleted. Wrapped (with LMK2 key). Encapsulated. KCV calculated. PIN generation/verification.	No
5[1]	CVV Verification Keys	Transported in shares.	Yes
		Generated. Deleted. Wrapped (with LMK2 key). Encapsulated. KCV calculated. CSS/CVC/CVV calculation/verification.	No
6[1]	EMV Authorization Keys	Transported in shares. Wrapped (with LMK2 key).	Yes
		Generated. Deleted. Encapsulated. Diversified. KCV calculated. EMV authorization transaction. Script security treatment.	No

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TDES3 LMKs definition (continued)

LMK	Description	Operation	Authorization Mode Only
7	Data Encryption Keys	Transported in shares.	Yes
		Generated. Deleted. Wrapped (with LMK2 key). Encapsulated. Diversified. KCV calculated. Data encryption/decryption.	No
8	MAC Keys	Transported in shares.	Yes
		Generated. Deleted. Transported in shares. Wrapped (with LMK2 key). Encapsulated. Diversified. KCV calculated. MAC Generation/Verification.	No
9[1]	Cash Keys	Transported in shares.	Yes
		Generated. Deleted. Wrapped (with LMK2 key). Encapsulated. Diversified. KCV calculated. Cash back certificate generation. Transport certificate generation.	No
10[1]	EMV Personalization Keys	Transported in shares. Wrapped (with LMK2 key).	Yes
		Generated. Deleted. Encapsulated. Diversified. KCV calculated.	No
11	TAF Keys	Generated. Deleted. Encapsulated. KCV calculated.	No
12	VIAT Keys	Encapsulated. KCV calculated.	No
13	Irreversible PIN Keys	Generated. Deleted. Transported in shares. Encapsulated. KCV calculated.	No

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TDES3 LMKs definition (continued)

LMK	Description	Operation	Authorization Mode Only
14[1]	Terminal Authentication Keys	Encapsulated.	Yes
		Generated. Deleted. Transported in shares. KCV calculated.	No
15[1]	Decimalization Tables	Generated. Deleted. Transported in shares. Wrapped (with LMK2 key). Encapsulated. KCV calculated. Decimalization table encryption.	No
16	Log Signature Keys	Generated. Wrapped (with LMK2 key). Log signature.	No
17	Password Encryption Keys	Transported in shares.	Yes
		Generated. Deleted. Wrapped (with LMK2 key). KCV calculated. Password encryption/decryption.	No
18	Deprecated	N/A	N/A
19[1]	Derivation Base Keys	Transported in shares.	Yes
		Generated. Deleted. Wrapped (with LMK2 key). Encapsulated. Diversified. KCV calculated.	No
20..26	Reserved	N/A	N/A
27	TR-31 Protection Keys	Transported in shares.	Yes
		Generated. Deleted. Wrapped (with LMK2 key). Encapsulated. KCV calculated. TR-31 Key Block protection.	No
28	IPEK Keys	Transported in shares.	Yes
		Diversified. Deleted. Wrapped (with LMK2 key). Encapsulated. KCV calculated.	No

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TDES3 LMKs definition (continued)

LMK	Description	Operation	Authorization Mode Only
29[1]	EMV Application cryptograms Keys	Transported in shares. Wrapped (with LMK2 key).	Yes
		Generated. Deleted. Encapsulated. Diversified. KCV calculated. EMV authorization transaction. Script security treatment.	No
30[1]	EMV Secure Messaging for Confidentiality Keys	Transported in shares. Wrapped (with LMK2 key).	Yes
		Generated. Deleted. Encapsulated. Diversified. KCV calculated. EMV authorization transaction. Script security treatment.	No
31[1]	EMV Secure Messaging for Integrity Keys	Transported in shares. Wrapped (with LMK2 key).	Yes
		Generated. Deleted. Encapsulated. Diversified. KCV calculated. EMV authorization transaction. Script security treatment.	No
32[1]	EMV Data Authentication Code Keys	Transported in shares. Wrapped (with LMK2 key).	Yes
		Generated. Deleted. Encapsulated. Diversified. KCV calculated. EMV authorization transaction. Script security treatment.	No
33[1]	EMV Dynamic Numbers Keys	Transported in shares. Wrapped (with LMK2 key).	Yes
		Generated. Deleted. Encapsulated. Diversified. KCV calculated. EMV authorization transaction. Script security treatment.	No

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TDES3 LMKs definition (continued)

LMK	Description	Operation	Authorization Mode Only
34, 35	Reserved	N/A	N/A
36	ISO 16609 MAC algorithm 1 (using TDEA) Keys	Transported in shares.	Yes
		Generated. Deleted. Transported in shares. Wrapped (with LMK2 key). Encapsulated. Diversified. KCV calculated. MAC Generation/Verification.	No
37	ISO 9797-1 MAC Algorithm 1 Keys	Transported in shares.	Yes
		Generated. Deleted. Transported in shares. Wrapped (with LMK2 key). Encapsulated. Diversified. KCV calculated. MAC Generation/Verification.	No
38	Reserved	N/A	N/A
39	ISO 9797-1 MAC Algorithm 3 Keys	Transported in shares.	Yes
		Generated. Deleted. Transported in shares. Wrapped (with LMK2 key). Encapsulated. Diversified. KCV calculated. MAC Generation/Verification.	No
40, 41	Reserved	N/A	N/A
42	ISO 9797-1:2011 MAC Algorithm 5/CMAC Keys	Transported in shares.	Yes
		Generated. Deleted. Transported in shares. Wrapped (with LMK2 key). Encapsulated. Diversified. KCV calculated. MAC Generation/Verification.	No

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TDES3 LMKs definition (continued)

LMK	Description	Operation	Authorization Mode Only
43	HMAC Keys	Transported in shares.	Yes
		Generated. Deleted. Transported in shares. Wrapped (with LMK2 key). Encapsulated. Diversified. KCV calculated. MAC Generation/Verification.	No
44	Reserved	N/A	N/A
45[1]	PIN Verification KPV other algorithm Keys	Transported in shares.	Yes
		Generated. Deleted. Wrapped (with LMK2 key). Encapsulated. KCV calculated. PIN generation/verification.	No
46[1]	PIN Verification IBM 3624 Keys	Transported in shares.	Yes
		Generated. Deleted. Wrapped (with LMK2 key). Encapsulated. KCV calculated. PIN generation/verification.	No
47[1]	PIN Verification VISA PVV Keys	Transported in shares.	Yes
		Generated. Deleted. Wrapped (with LMK2 key). Encapsulated. KCV calculated. PIN generation/verification.	No
48, 49	Reserved	N/A	N/A
50[1]	PIN Block Protection Key	KCV calculated. PIN block LMK encryption.	No
51..59	Reserved	N/A	N/A
60	Asymmetric Encryption Keys	LMK Change.[2]	Yes
		Generated. Wrapped (with LMK2 key). Data encryption/decryption.	No
61	Asymmetric Signature Keys	LMK Change.[2]	Yes
		Generated. Wrapped (with LMK2 key). Signature generation/verification.	No

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TDES3 LMKs definition (continued)

LMK	Description	Operation	Authorization Mode Only
62[2]	Asymmetric Encryption Signature Keys	LMK Change.	Yes
		Generated. Wrapped (with LMK2 key). Data encryption/decryption. Signature generation/verification.	No
63	Transport Signature Keys	Generated. Transport signature generation/verification.	No
64..99	Reserved	N/A	N/A

[1] Cryptosec-Banking Only.
[2] Not allowed on PCI PTS HSM v4.0-Approved mode.

1.3 AES-256 LMKs

The AES-256 Local Master Keys are AES-256 keys derived from the AES-256 Master Key. The AES-256 Master Key is loaded in shares through the console interface.

Keys stored in databases will be associated with one LMK index. Thus, the keys are grouped by their purpose. Grouping the keys ensures that a key is not used for anything else than its intended purpose.

Table 1.2: AES-256 LMKs definition

LMK	Description	Operation	Authorization Mode Only
0	AES-256 Master Key	KCV calculated.	No
1	Key Block Confidentiality Key	Key block LMK.	N/A
2	Key Block Authenticity Key	Key block LMK.	N/A
3[1]	PIN Block Confidentiality Key	PIN block LMK.	N/A

[1] Cryptosec-Banking Only.

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Chapter 2

HSM Communication Protocol

2.1 Data Types

The HSM receives and emits data. Each data will belong to one of the following data types.

Table 2.1: Data types

Type	Description
A	Alphanumeric.
H	Hexadecimal.
N	Numeric.
B	Binary.
K	Key Container.
V	RSA Keys Data Base.
T	TR-31 Key Block.

2.1.1 Basic Data Types

The basic data types are formed by arrays of data of the same nature:

A array of alphanumeric characters. Each element is expressed as one of the following ASCII values:

- 0..9.
- A..Z.
- a..z.

H array of hexadecimal characters. Each element is expressed as one of the following ASCII values:

- 0..9.
- A..F.

- a..f.

N array of decimal characters. Each element is expressed as one of the following ASCII values:

- 0..9.

B array of bytes. Each element is expressed as two hexadecimal characters.

An array length is defined in bytes. A length can be:

Fixed when a fixed length value is specified.

Configured when the length value is configured by other means, i.e. through console.

Variable when the length value is provided as a separate numeric data or is obtainable by other means.

2.1.2 Complex Data Types

The complex data types are formed by combinations of basic data.

2.1.2.1 Container Version 0

The container version 0 is used to store keys.

They are encrypted under an LMK using TDES3 algorithm.

Table 2.2: Container version 0 structure

Type	Length	Description
A	1	Type of key: **D** DES keys for external storage. **A** AES keys for external storage.[1] **R** RSA private keys for external storage. **I** IBM format RSA private keys structured in CRT.[2] **S** DES keys stored internally. **T** AES keys stored internally.[2] **P** RSA public keys for external storage.
N	4	Key value length.[3]
H	var	Key value.[4]
A	3	Key encryption.[5]
N	2	KCV value length.
H	var	KCV value.[6]

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Container version 0 structure (continued)

Type	Length	Description

[1] Encryption with an LMK deprecated.
[2] Deprecated.
[3] When key type is **I** then key length shall be 5000 bytes.
[4] Dependent on the key type, it will appear in the following format:

DES Keys stored externally or to be stored internally, in hexadecimal format.

DES Key stored internally, five digits corresponding to the key identifier returned by the HSM.

RSA Public Keys, its DER codification in the structure ASN.1 defined in the PKCS#1 standard.

RSA Private Keys (not in IBM format structured in CRT), whose value is the DER codification in the structure ASN.1 PrivateKeyInfo defined in the standard PKCS#8.

RSA Private Keys in IBM format structured in CRT: IBM proprietary structure.

[5] Type encryption key:

L00 .. L99 indicate that the key is encrypted with an LMK.

D00 indicates that the key is encrypted with DES algorithm, with an external key.

A00 indicates that the key is encrypted with AES algorithm, with an external key.

N00 indicates DES key internally stored. Also indicates that the key is not encrypted.

R00 indicates that the key is encrypted with RSA algorithm, with an external key.

[6] For DES and AES Keys, this field represents its KCV. For RSA public keys, this field represents its MAC.
KCV of the key will be provided when the key type supports it and it is wished to specify another key verification system:

For DES Keys, if the KCV will be not specified, this field shall be set to '00' value.

For AES Keys, this field shall be specified.

For RSA private keys, this field shall be empty.

For more details please refer to [Container Format, Container format for version 0].

Note: This is a legacy container.

2.1.2.2 Container Version 1

The container version 1 is used to store keys and decimalization tables.

The data and some of its properties like its purpose are AES encrypted for confidentiality and HMACed for integrity and authentication.

Table 2.3: Container version 1 structure

Type	Length	Description
A	1	**Z**.
N	6	Data value length.
H	var	Data value.

This format is mandatory if:

- The HSM is in PCI PTS HSM v4.0-Approved mode and AES Keys are needed.

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- The HSM is in NIST FIPS PUB 140-2-Approved mode.

For more details please refer to [Container Format, Container format for version 1].

2.1.2.3 RSA Keys Data Base

It is possible to store RSA private or public keys in a data base.

The data base objects associate the key container, regardless of its version, with an alphanumeric label.

Table 2.4: RSA keys data base structure

Type	Length	Description
A	1	Type of key: **V** Internal data base RSA key storage.
N	4	Key label length.[1]
A	var	Key label.
A	3	**N00**

[1] Up to 49 alphanumeric characters.

2.1.2.4 TR-31 Key Blocks

The Key Block with Optional Block defined in [X9.TR-31, Annex A] is also supported. Nevertheless, only the following Key Block Version IDs are accepted:

B key block protected using the TDEA Key Derivation Binding Method.

D key block protected using the AES Key Derivation Binding Method.

The Key Block Version ID is the first character in the block.

Table 2.5: TR-31 Key Block structure

Type	Length	Description
A	var	TR-31 Key Block.

The length of the block is part of the block itself as four ASCII numeric digits from the second character in the block. Therefore is not specified separately.

For more details please refer to [X9.TR-31].

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2.2 Command Identifiers

The Command Identifier is a fixed length four byte hexadecimal value.

It names a command. It is provided for each command defined in Host Commands.

2.3 Error Code

An Error Code is a fixed length eight byte hexadecimal value.

It summarizes a command execution result. The possible values are defined in Error Codes.

2.4 Message Body

A message body is formed by one or several concatenated data.

There are two types of message bodies.

2.4.1 Request Body

It contains the requested operation, along with the associated input data if needed.

Table 2.6: Request body

Type	Length	Description
H	4	Command Identifier.
?[1]	?[2]	Request data 1.[3]
..		
?[1]	?[2]	Request data N.[3]

[1] One of the defined data types.
[2] A valid length for the data type.
[3] Request data number, data type and length depend on the specific command.

2.4.2 Response Body

It contains the result of the requested operation, along with the associated output data if they exist.

Table 2.7: Response body

Type	Length	Description
H	4	Command Identifier.[1]
H	8	Error code.

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Response body (continued)

Type	Length	Description
?[2]	?[3]	Response data 1.[4,5]
..		
?[2]	?[3]	Response data N.[4,5]

[1] Same value as in the request body.
[2] One of the defined data types.
[3] A valid length for the data type.
[4] Response data may be present only if the command has succeeded.
[5] Response data number, data type and length depend on the specific command.

2.5 Command Header

The command header is a configured length alphanumeric data. Its length is up to 199 characters and it is configured by the Administrator through the HSM console.

The command header is provided in the request message and returned unchanged in the response message. It can be used for command identification.

2.6 Message

A message is formed by the concatenation of three fields.

Table 2.8: Message structure

Type	Length	Description
D	6	Next fields length.[1]
A	HL	Command Header.
-	BL	Message Body.

[1] Its value is HL + BL. Up to 999999.

There are two types of messages:

Request Message when the message body is a request body.

Response Message when the message body is a response body.

2.7 Command

A command is formed by:

- A request message sent from the Host to the HSM.
- A response message sent from the HSM to the Host.

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Chapter 3

Host Commands

When the HSM is configured in the PCI PTS HSM v4.0 or the NIST FIPS PUB 140-2 approved modes, refer to their respective Non-Proprietary Security Policies to get further information regarding the restrictions derived from working in such modes.

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3.1 Key Management

3.1.1 0101 - Key Generation

Generate any key of the supported types.

Request Body:

Data	Length & Type	Description
Command Identifier	4H	**0101**.
LMK	2N	LMK label associated with the key to generate. The value shall be different from 00. ***Value according to Local Master Keys.***
Key Type	1A	Type of the key to generate: **D** - DES keys for external storage in v0 container. **S** - DES keys for internal storage in v0 container. **E** - DES keys for external storage in v1 container. **U** - DES keys for internal storage in v1 container. **B** - AES keys for external storage in v1 container. **W** - AES keys for internal storage in v1 container. **L** - GENERIC keys for external storage in v0 container. **M** - GENERIC keys for external storage in v1 container. **R** - RSA private keys for external storage in v0 container. **X** - RSA private keys for external storage in v1 container.
Key Length	4N	Length in bits of the key to generate. Dependent on **Key Type**, it will take the following values: - **D**, **S**, **E**, **U**: **0064**, **0128** or **0192**. - **B**, **W**: **0128**, **0192** or **0256**. - **R**, **X**: from **0512** to **4096** in 64 bits steps.
Public Key Exponent	1N	Exponent of the RSA key to generate: **0** - 3. **1** - Fermat4. ***Only if Key Type is R or X.***
Key Generation Mode	2N	RSA key generation Miller-Rabin test number of rounds: **01** - Automatically generated based on **L** field for less than 2^{-100} false positive probability. **02..50** - As indicated by this value. ***Only if Key Type is R or X.***

Response Body:

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Data	Length & Type	Description
Command Identifier	4H	**0101**.
Error Code	8H	Error code received from the HSM: **00000000** - No error code. **xxxxxxxx** - Other code defined in Error Codes.
Generated Key	varK	The secret or private key under the selected **LMK**.
RSA Public Key	varK	The RSA public key under the selected **LMK**. ***Only if Key Type is R or X.***

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3.1.2 0102 - Key Deletion

Delete a DES or AES key stored internally.

Request Body:

Data	Length & Type	Description
Command Identifier	4H	**0102**.
Key	varK	Internal key to delete.

Response Body:

Data	Length & Type	Description
Command Identifier	4H	**0102**.
Error Code	8H	Error code received from the HSM: **00000000** - No error code. **xxxxxxxx** - Other code defined in Error Codes.

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3.1.3 0103 - Keys Listing

Lists the DES or AES keys stored internally.

Request Body:

Data	Length & Type	Description
Command Identifier	4H	**0103**.
Key Type	1A	Optional. Type of keys to list: **S** - DES keys. **T** - AES keys. ***If not provided, DES keys will be listed.***

Response Body:

Data	Length & Type	Description
Command Identifier	4H	**0103**.
Error Code	8H	Error code received from the HSM: **00000000** - No error code. **xxxxxxxx** - Other code defined in Error Codes.
Number of Keys	5N	Number of internal keys. ***This value depends on the HSM state.***
Key Index #n	5N	Internal key index.[1]
Key LMK #n	2N	LMK that protects the key.[1]

[1] The greyed fields will be repeated as many times as specified on field **Number of Keys**.

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3.1.4 0104 - Delete Internal RSA Key

Delete an internal RSA key stored internally.

Request Body:

Data	Length & Type	Description
Command Identifier	4H	**0104**.
RSA Label Length	2N	Next field length.
RSA Label	varA	The RSA key label.

Response Body:

Data	Length & Type	Description
Command Identifier	4H	**0104**.
Error Code	8H	Error code received from the HSM: **00000000** - No error code. **xxxxxxxx** - Other code defined in Error Codes.

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3.1.5 0105 - List Internal RSA Keys

Get a list with all the RSA key labels which are store internally.

Request Body:

Data	Length & Type	Description
Command Identifier	4H	**0105**.

Response Body:

Data	Length & Type	Description
Command Identifier	4H	**0105**.
Error Code	8H	Error code received from the HSM: **00000000** - No error code. **xxxxxxxx** - Other code defined in Error Codes.
Number of Keys	8N	Number of internal keys. ***This value depends on the HSM state.***
RSA Label Length	2N	Next field length.[1]
RSA Label	varA	The RSA key label.[1]

[1] The greyed fields will be repeated as many times as specified on field **Number of Keys**.

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3.1.6 0106 - Get Internal RSA Key

Get the cryptogram of any internal RSA key.

Request Body:

Data	Length & Type	Description
Command Identifier	4H	**0106**.
RSA Label Length	2N	Next field length.
RSA Label	varA	The RSA key label to retrieve.

Response Body:

Data	Length & Type	Description
Command Identifier	4H	**0106**.
Error Code	8H	Error code received from the HSM: **00000000** - No error code. **xxxxxxxx** - Other code defined in Error Codes.
RSA Key	varK	The retrieved RSA key.

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3.1.7 0107 - Store RSA Key

Store an RSA Key into the HSM given a RSA Key cryptogram.

Request Body:

Data	Length & Type	Description
Command Identifier	4H	**0107**.
RSA Key	varK	The RSA key to store.
RSA Label Length	2N	Next field length.
RSA Label	varA	Label under which it is going to be stored.

Response Body:

Data	Length & Type	Description
Command Identifier	4H	**0107**.
Error Code	8H	Error code received from the HSM: **00000000** - No error code. **xxxxxxxx** - Other code defined in Error Codes.

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3.1.8 0108 - Save Asn1 RSA Public Key

Save a RSA public key in any Cryptosec Dekaton container into Asn1 DER format.

Request Body:

Data	Length & Type	Description
Command Identifier	4H	**0108**.
Public Key encoding	2H	Encoding rules for public key: **0** - DER encoding for ASN.1 Public Key (INTEGER uses unsigned representation). **1** - DER encoding for ASN.1 Public Key (INTEGER uses 2's complement representation).
Public Key	varK	Key to get its Asn1 representation.

Response Body:

Data	Length & Type	Description
Command Identifier	4H	**0108**.
Error Code	8H	Error code received from the HSM: **00000000** - No error code. **xxxxxxxx** - Other code defined in Error Codes.
Public Key Length	6N	Next field length.
Public Key	varH	Asn1 Public Key.

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3.1.9 0202 - Key Import

Import an external key which it is encrypted under a transport key.

Request Body:

Data	Length & Type	Description
Command Identifier	4H	**0202**.
Key Type	2N	LMK index for the imported key. The value should be different from 00. ***Value according to Local Master Keys.***
Algorithm	2H	Import Algorithm: **00** - ECB. **01** - RSA-PKCS#1 v1.5. **02** - RSA-OAEP. **03** - CBC.
Key to import	varK	Key to import encrypted under the **Transport Key**.
Transport Key	varK	Decryption key (under LMK 02, LMK 60 or LMK 62). ***LMK60 and LMK62 only allowed when the HSM is not configured in PCI PTS HSM v4.0-Approved mode.***
Store Type	1A	Type of the key to import: **D** - DES keys for external storage in v0 container. **S** - DES keys for internal storage in v0 container. **E** - DES keys for external storage in v1 container. **U** - DES keys for internal storage in v1 container. **B** - AES keys for external storage in v1 container. **W** - AES keys for internal storage in v1 container. **L** - GENERIC keys for external storage in v0 container. **M** - GENERIC keys for external storage in v1 container. **R** - RSA private keys for external storage in v0 container. **X** - RSA private keys for external storage in v1 container. ***Optional. Default value: D.***
Initialization Vector	16/32H[1]	Initialization vector. ***Only if Algorithm is 03.***
Delimiter	1A	**;** (0x3B ASCII). Optional. If present, **Mask Generation Function** field shall exist. ***Only if Algorithm is 02. If MGF1 will not be informed, SHA-1 will be used.***

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Data	Length & Type	Description
Mask Generation Function	1A	MGF1 hash function: **0** - SHA-1 **1** - SHA-224 **2** - SHA-256 **3** - SHA-384 **4** - SHA-512 **5** - SHA-512/224 **6** - SHA-512/256 ***Only if Algorithm is 02.***
Delimiter	1A	**#** (0x23 ASCII). Optional. If present, **Wrapping Method** field shall exist. ***Only if the HSM is configured in Non-Approved mode and Algorithm is 01 or 02.***
Wrapping Method	1N	Wrapping method: **0** - PCI PTS HSM v4.0-Approved. **1** - NIST FIPS PUB 140-2-Approved. **2** - Non-Approved (Legacy). ***Only if the HSM is configured in Non-Approved mode and Algorithm is 01 or 02.***
Verification Key	varK	RSA key for signature verification (under LMK63). ***Only if Algorithm is 01 or 02 and the Wrapping Method is 0.***

[1] 16 for DES keys and 32 for AES keys.

Response Body:

Data	Length & Type	Description
Command Identifier	4H	**0202**.
Error Code	8H	Error code received from the HSM: **00000000** - No error code. **xxxxxxxx** - Other code defined in Error Codes.
Imported Key	varK	The imported secret/private key under the selected **Key Type**.
Imported Public Key	varK	The imported Public key under the selected **Key Type**. ***Only if the imported key is RSA.***

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3.1.10 0203 - Import Signed RSA Public Key

Import an external signed RSA public keys into the HSM.

Request Body:

Data	Length & Type	Description
Command Identifier	4H	**0203**.
Key Type	2N	LMK index for the imported key. Supported values 60, 61 or 62. ***LMK62 only allowed when the HSM is not configured in PCI PTS HSM v4.0-Approved mode.***
Key Container Type	1N	Public key format: **0** - Public key ASN.1 structure. **1** - X509v3 certificate.
Public key Container Length	4N	Next field length. ***Only if Key Container Type is 1.***
Public key Container	varH	Public key container. Public Key or certificate, depending on the Key container Type selected.
Signature Length	4N	Next field length.
Signature	varH	The signature of the Key to import.
Verification Key	varK	The private key to verify the previous signature (under LMK61 or LMK62). ***LMK62 only allowed when the HSM is not configured in PCI PTS HSM v4.0-Approved mode.***
Storage Type	1A	Storage type: **P** - RSA public keys within container v0. **Y** - RSA public keys within container v1. ***Optional. Default value: P.***

Response Body:

Data	Length & Type	Description
Command Identifier	4H	**0203**.
Error Code	8H	Error code received from the HSM: **00000000** - No error code. **xxxxxxxx** - Other code defined in Error Codes.
Imported Key	varK	The imported key under the selected **Key Type**.

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3.1.11 0204 - Import Unsigned RSA Public Key

Import external RSA public keys without the signature of another RSA key into the HSM.

The method can behave in two modes:

Bulk mode where the HSM automatically imports the key without human intervention. This mode is subject to the following conditions:

- Bulk mode is enabled.

  Note: In PCI PTS HSM v4.0-Approved mode, bulk mode is automatically disabled after 24 hours or after 8640 successful bulk imports.

Request mode where the HSM waits for Administrator approval to import the key. This mode is subject to the following conditions:

- Bulk mode is not enabled.
- The HSM is configured to allow an import request. This is detailed in [Administrator Guide, RSA Public Key Import Request Management].

  Note: The allowance is configured from an Administrator session. It is recommended for operative and security reasons to allow an import request only as part of an ongoing import operation. Once a key is imported the allowance is automatically cleared, implying that each key import operation needs to be explicitly allowed.
- The Administrator checks that the SHA-512 digest value of the ASN.1 encoding of the key to import presented through console is correct; and based on that authorizes the key import.

  Note: While it is not required that an Administrator session is opened when the command is issued, the digest checking and import authorization are to be done from such session. The key import operation stalls until it is explicitly accepted or rejected.

The mode selection is done through HSM configuration in an Administrator session. This is detailed in [Administrator Guide, RSA Public Key Import Management].

Note: Request mode is selected by default for backwards compatibility and security reasons.

If bulk mode is not enabled and an import request is not allowed no mode is selected. Error code **0000E000** is returned.

Request Body:

Data	Length & Type	Description
Command Identifier	4H	**0204**.
Key Type	2N	LMK index for the imported key. Supported values 60, 61 or 62. ***LMK62 only allowed when the HSM is not configured in PCI PTS HSM v4.0-Approved mode.***
Public Key	varK	Public key to import in ASN.1 format.

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Data	Length & Type	Description
Storage Type	1A	Storage type: **P** - RSA public keys within container v0. **Y** - RSA public keys within container v1. ***Optional. Default value: P.***

Response Body:

Data	Length & Type	Description
Command Identifier	4H	**0204**.
Error Code	8H	Error code received from the HSM: **00000000** - No error code. **xxxxxxxx** - Other code defined in Error Codes.
Imported Key	varK	The imported key under the selected **Key Type**.

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3.1.12 0206 - Key Import in TR-31 Format (with proposed LMK)

Import an external key which it is encrypted according to TR-31 standard into the HSM.

Request Body:

Data	Length & Type	Description
Command Identifier	4H	**0206**.
Key Storage Type	1N	Key storage: **0** - External. **1** - Internal.
LMK	2N	LMK label associated with the key to import. The value shall be different from 00. ***Value according to Local Master Keys.***
Key Container	1N	Key container version: **0** - v0 container. **1** - v1 container.
Transport Key	varK	Decrytion key (under LMK2 (not allowed in PCI PTS HSM v4.0-Approved mode) or LMK27).
Key	varT	Key to be imported encrypted according to TR-31 standard.[1]

[1] Please see [X9.TR-31, Annex A] and [TR-31 Implementation].

Response Body:

Data	Length & Type	Description
Command Identifier	4H	**0206**.
Error Code	8H	Error code received from the HSM: **00000000** - No error code. **xxxxxxxx** - Other code defined in Error Codes.
Imported Key	varK	The imported key.

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3.1.13 0207 - Key Import in AESGCM-KW Format

Import an AES data encryption key according to the AESGCM-KW (RFC 7518) standard.

Request Body:

Data	Length & Type	Description
Command Identifier	4H	**0207**.
Key Storage Type	1N	Key storage: **0** - External. **1** - Internal.
Key Container	1N	Key container version: **1** - v1 container.
Transport Key	varK	The import key (under LMK2).
Cryptogram Length	2N	Next field length.
Cryptogram	varH	Cryptogram of the key to import.
IV	24H	Initialization vector.
Tag	32H	Tag.

Response Body:

Data	Length & Type	Description
Command Identifier	4H	**0207**.
Error Code	8H	Error code received from the HSM: **00000000** - No error code. **xxxxxxxx** - Other code defined in Error Codes.
Imported Key	varK	The imported key (under LMK7).

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3.1.14 0301 - Export Symmetric Key in Shares

Export a DES or AES key in components through console. It can also generate the key to export.

Request Body:

Data	Length & Type	Description
Command Identifier	4H	**0301**.
Mode	1H	Key component generation mode: **0** - Generates key and exports it in components. **1** - Exports key included in the message.
Key Type	2N	LMK for the generated key. ***Value according to Local Master Keys.*** ***Only if Mode is 0.***
Key Length	2N	Length of the key to generate (in bytes). ***Only if Mode is 0.***
Components Number	1N	Number of shares to return (min. **2** - max. **9**).
Key To Export	varK	The key to be exported. ***Only if Mode is 1.***
User Data Length	4N	Next field length.
User Data	varA	Table of variable user data (name, custodian, ATM, dates, finality, key, etc.). Format depends on the definition of the printing format.
Key Storage Type	1A	Storage type: **D** - DES keys for external storage in v0 container. **S** - DES keys for internal storage in v0 container. **E** - DES keys for external storage in v1 container. **U** - DES keys for internal storage in v1 container. **B** - AES keys for external storage in v1 container. **W** - AES keys for internal storage in v1 container. **L** - GENERIC keys for external storage in v0 container. **M** - GENERIC keys for external storage in v1 container. ***Only if Mode is 0.*** ***Optional. Default value: D.***

Response Body:

Data	Length & Type	Description
Command Identifier	4H	**0301**.

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Data	Length & Type	Description
Error Code	8H	Error code received from the HSM: **00000000** - No error code. **xxxxxxxx** - Other code defined in Error Codes.
Generated Key	varK	The generated key under the selected **Key Type**. ***Only if Mode is 0.***

3.1.14.1 Key Printing

It is necessary to define the printing template by running the command 1301 before performing printing operation.

In this command, the argument T contains print strings 0 to 15 that are referenced in the template. There is no need to define all of the 16 strings. However, it shall be taken into account that they are numbered in a sequential manner starting from 0. Sequence `\0` is used as a string delimiter. The sequence `^P` induces the printing of the key component. Sequence `^T` involves printing the KCV Key component.

As an example, consider the generation and two components printing of a key, together with its KCV for control purposes. It would be done as follows:

```
COMPONENT: XXXXXXXXXXXXXXXX KCV: YYYYYY
```

Where `X..X` is the component and `Y..Y` is the KCV. In this case the template is kept to a minimum:

```
>L^0^P^1^T>F
```

Where the sequence `^0` denotes the inclusion of the first element on the data table (shall be `\COMPONENT: "`) and the sequence `^1` denotes the point of including the second element of the data table (shall be `\KCV: "`). `^P` denotes printing the key component and `^T` denotes printing the component KCV.

Therefore, with this template, to obtain the wanted printing, the data table to include in the command is:

```
COMPONENT: \0 KCV:
```

The `\0` is used as separator, and it is not necessary to include it following the last string. This string has a length of 00000020.

The alternative to simplify (in this case, eliminate) the data table would be including data strings into the templates:

```
>L COMPONENT: ^P KCV: ^T>F
```

Although this approach is less proper from a formal perspective.

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3.1.15 0303 - Key Export

Export a symmetric key using a transport key.

For security reasons, if Wrapping Method is 2, one of the data to append or prepend shall exist and the addition of their lengths shall be multiple of 16 hexadecimal digits.

Request Body:

Data	Length & Type	Description
Command Identifier	4H	**0303**.
Algorithm	2H	Export algorithm: **00** - ECB. **01** - RSA-PKCS#1 v1.5. **02** - RSA-OAEP. **03** - CBC.
Key To Export	varK	The key to be exported.
Prepend Data Length	4N	Next field length. ***If Wrapping Method is 0, this field shall be 0000.*** ***If Wrapping Method is 2, Prepend Data Length or Append Data Length shall be different from 0000, besides if Algorithm is 00 or 03, the addition of both of them shall be multiple of 16 hexadecimal digits.***
Prepend Data	varH	Data to prepend to the key before it is encrypted.
Transport Key	varK	The export key (under LMK02, LMK60 or LMK62). ***LMK60 and LMK62 only allowed when the HSM is not configured in PCI PTS HSM v4.0-Approved mode.***
Initialization Vector	16/32H[1]	The initialization vector. ***Only if Algorithm is 03.***
Delimiter	1A	**;** (0x3B ASCII). Optional. If present, **Mask Generation Function** field shall exist. ***Only if Algorithm is 02. If MGF1 will not be informed, SHA-1 will be used.***
Mask Generation Function	1A	MGF1 hash function: **0** - SHA-1 **1** - SHA-224 **2** - SHA-256 **3** - SHA-384 **4** - SHA-512 **5** - SHA-512/224 **6** - SHA-512/256 ***Only if Algorithm is 02.*** ***Shall exist if the above delimiter is present.***

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Data	Length & Type	Description
Delimiter	1A	**#** (0x23 ASCII). Optional. If present, **Wrapping Method** and **Append Data Length** fields shall exist.
Wrapping Method	1N	Wrapping method: **0** - PCI PTS HSM v4.0-Approved. **1** - NIST FIPS PUB 140-2-Approved. **2** - Non-Approved (Legacy). ***Shall exist if the above delimiter is present.***
Append Data Length	4N	Next field length. ***If Wrapping Method is 0, this field shall be 0000.*** ***If Wrapping Method is 2, Prepend Data Length or Append Data Length shall be different from 0000, besides if Algorithm is 00 or 03, the addition of both of them shall be multiple of 16 bytes.***
Append Data	varH	Data to append to the key before it is encrypted.
Signature Key	varK	RSA key to sign the export cryptogram (under LMK63). ***Only if Algorithm is 01 or 02 and the Wrapping Method is 0.***

[1] 16 for DES keys and 32 for AES keys.

Response Body:

Data	Length & Type	Description
Command Identifier	4H	**0303**.
Error Code	8H	Error code received from the HSM: **00000000** - No error code. **xxxxxxxx** - Other code defined in Error Codes.
Exported Key	varK	The exported key encrypted under the **Transport Key**.

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3.1.16 0304 - Key Export in TR-31 Format

Export a key according to TR-31 standard.

Request Body:

Data	Length & Type	Description
Command Identifier	4H	**0304**.
Key To Export	varK	The key to be exported.
Transport Key	varK	The export key (under LMK2 (not allowed in PCI PTS HSM v4.0-Approved mode) or LMK27).
KBH Information	8A	The Key Block Header in the following order: KeyBlockVersionID || KeyUsage || Algorithm || ModeOfUse || KeyVersionNumber || Exportability.[1]
Number of Optional Blocks	2N	Number of optional blocks.[1]
Optional Blocks Length	4N	Next field length.[1] ***Only if Number Optional Block is not 00.***
Optional Blocks	varH	TR-31 Optional blocks to include.[1] ***Only if Number Optional Block is not 00.***

[1] Please see [X9.TR-31, Annex A] and [TR-31 Implementation].

Response Body:

Data	Length & Type	Description
Command Identifier	4H	**0304**.
Error Code	8H	Error code received from the HSM: **00000000** - No error code. **xxxxxxxx** - Other code defined in Error Codes.
Exported Key	varT	The exported key according to TR-31 standard.

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3.1.17 0305 - Key Export in AESGCM-KW Format

Export an AES data encryption key according to the AESGCM-KW (RFC 7518) standard.

Request Body:

Data	Length & Type	Description
Command Identifier	4H	**0305**.
Key To Export	varK	The key to be exported (under LMK7).
Transport Key	varK	The export key (under LMK2).
Input IV Length	2N	Next field length. ***Shall be 24 if the IV is provided (not allowed in NIST FIPS PUB 140-2-Approved mode) or 00 to internally generate the IV.***
Input IV	varH	Initialization vector. ***Only if IV Length is not 00.***

Response Body:

Data	Length & Type	Description
Command Identifier	4H	**0305**.
Error Code	8H	Error code received from the HSM: **00000000** - No error code. **xxxxxxxx** - Other code defined in Error Codes.
Cryptogram Length	2N	Next field length.
Cryptogram	varH	Cryptogram of the exported key.
Output IV	24H	Initialization vector for the import.
Tag	32H	Tag for the import.

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3.1.18 0402 - TDES3 Master Key and LMK KCV Calculation

Computes the KCV of the TDES3 Master Key or the TDES3 LMK according to [X9.24-1] using the Legacy KCV computation method.

Request Body:

Data	Length & Type	Description
Command Identifier	4H	**0402**.
LMK	2N	Index of the LMK.[1] ***If LMK is 00 the KCV of the TDES3 Master Key is computed.***

[1] Value according to Local Master Keys.

Response Body:

Data	Length & Type	Description
Command Identifier	4H	**0402**.
Error Code	8H	Error code received from the HSM: **00000000** - No error code. **xxxxxxxx** - Other code defined in Error Codes.
KCV	6H	Computed KCV.

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3.1.19 0403 - AES Master Key KCV Calculation

Compute the KCV of the AES-256 Master Key according to [X9.24-1] using the CMAC KCV computation method. Only the first six characters from the 10-character result are used.

Request Body:

Data	Length & Type	Description
Command Identifier	4H	**0403**.

Response Body:

Data	Length & Type	Description
Command Identifier	4H	**0403**.
Error Code	8H	Error code received from the HSM: **00000000** - No error code. **xxxxxxxx** - Other code defined in Error Codes.
KCV	6H	Computed KCV.[1]

[1] Only the first six characters from the 10-character result are used.

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3.1.20 0404 - Key KCV Calculation

Computes KCV of a symmetric key according to [X9.24-1]. The first six digits of the following operations are used as KCV:

- For DES keys the Legacy KCV computation method (encryption of a block of binary zeros) is used.
- For AES keys the CMAC KCV computation method (CMAC of a block of binary zeros) is used.
- For HMAC (generic) keys: HMAC of a zero-length message.

Request Body:

Data	Length & Type	Description
Command Identifier	4H	**0404**.
Key	varK	The key to get its check value.
Hash Function	2N	Hash function: **00** - MD5. **01** - SHA-1. **02** - SHA-224. **03** - SHA-256. **04** - SHA-384. **05** - SHA-512. **06** - SHA-512/224. **07** - SHA-512/256. ***Only if Key is an HMAC (generic) key.***

Response Body:

Data	Length & Type	Description
Command Identifier	4H	**0404**.
Error Code	8H	Error code received from the HSM: **00000000** - No error code. **xxxxxxxx** - Other code defined in Error Codes.
KCV Length	2N	Next field length. **06**.
KCV	varH	Computed KCV.

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3.1.21 1801 - Key Storage Update

Allows the re-encryption of a key, that is encrypted with the old LMK, to be encrypted under the new LMK.

Read next section to be aware of very important considerations regarding this command.

NOTICE: This command has to be used right after new Master Keys are loaded.

Request Body:

Data	Length & Type	Description
Command Identifier	4H	**1801**.
Key	varK	The key to be updated.

Response Body:

Data	Length & Type	Description
Command Identifier	4H	**1801**.
Error Code	8H	Error code received from the HSM: **00000000** - No error code. **xxxxxxxx** - Other code defined in Error Codes.
Updated Key	varK	The updated key.

3.1.21.1 Key Update

For this command to work properly, the following items shall be taken into account:

- Have loaded new Master Keys into the HSM.
- Have kept the old LMKs of the previous Master Keys.
- The key to be updated shall be encypted under an old LMK: the update process is continuous because only the most recently replaced set of LMKs is stored.
- Not having deleted the old LMKs prior to the execution of this command. Once removed, it will only be possible to recover any key encrypted under this set of keys loading again the associated Master Keys.

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3.1.22 1802 - Key Container Update

Allows the update of the container of a key to a higher container version.

Request Body:

Data	Length & Type	Description
Command Identifier	4H	**1802**.
Key	varK	The key to be updated.
Version	2N	Updated key container version.

Response Body:

Data	Length & Type	Description
Command Identifier	4H	**1802**.
Error Code	8H	Error code received from the HSM: **00000000** - No error code. **xxxxxxxx** - Other code defined in Error Codes.
Updated Key	varK	The updated key.

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3.2 EMV Transaction Authorization (Cryptosec-Banking Only)

3.2.1 0501 - Data Authentication Code Validation

Verify a Data Authentication Code.

Request Body:

Data	Length & Type	Description
Command Identifier	4H	**0501**.
Key	varK	DAC validation key (under LMK32 or LMK6).
Mandatory Legacy Field	1N	**1**.
PAN	16N	Pre-formatted PAN / PANseq No.
Data Authentication Code	4H	DAC to validate.

Response Body:

Data	Length & Type	Description
Command Identifier	4H	**0501**.
Error Code	8H	Error code received from the HSM: **00000000** - No error code. **xxxxxxxx** - Other code defined in Error Codes.
Validation Result	1N	Validation result: **0** - Correct. **1** - Failed.
Diagnostic DAC	4H	The calculated DAC value. ***Only if Validation Result is 1 and the HSM is in Authorised State.***

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3.2.2 0502 - Dynamic Number Validation

Verify a Dynamic Number.

Request Body:

Data	Length & Type	Description
Command Identifier	4H	**0502**.
Key	varK	DN validation key (under LMK33 or LMK6).
Mandatory Legacy Field	1N	**1**.
PAN	16N	Pre-formatted PAN / PANseq No.
Calculation Data	16H	ATC || 00 00 || UN. Usually UN=00 00 00 00.
Dynamic Number	4H	DN to validate.

Response Body:

Data	Length & Type	Description
Command Identifier	4H	**0502**.
Error Code	8H	Error code received from the HSM: **00000000** - No error code. **xxxxxxxx** - Other code defined in Error Codes.
Validation Result	1N	Validation result: **0** - Correct. **1** - Failed.
Diagnostic DN	4H	The calculated DN value. ***Only if Validation Result is 1 and the HSM is in Authorised State.***

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3.2.3 0503 - ARQC Verification and/or ARPC Generation (EMV 3.1.1)

Verify an ARQC and/or generate an ARPC. Diagnostic data is produced by this command only if the HSM is in Authorized State.

This command supports the following cryptogram methods:

Application	CVN
VISA VIS	10 or 18
MasterCard M/Chip	10 or 11

Request Body:

Data	Length & Type	Description
Command Identifier	4H	**0503**.
Mode	1N	Execution mode: **0** - Validate ARQC. **1** - Validate ARQC and calculate ARPC. **2** - Calculate ARPC.
Scheme	1N	The product: **0** - Visa. **1** - Mastercard.
MK-AC	varK	The Issuer Master Key for generating and verifying application cryptograms (under LMK29 or LMK6).
PAN / PANseq No	16H	Pre-formatted PAN / PANseq No.
ATC	4H	The Application Transaction Counter from the card. This is used for session key generation. ***Only if Scheme is 1 and Mode is NOT 2.***
UN	8H	The Unpredictable Number. This is used for session key generation. ***Only if Scheme is 1 and Mode is NOT 2.***
Transaction Data Length	3N	Next field length. ***Shall be even and not zero.*** ***Only if Mode is 0 or 1.***
Transaction Data	varH	Transaction data (do not pad). ***Only if Mode is 0 or 1.***
ARQC/TC/AAC	16H	ARQC/TC/AAC to validate and/or use to generate ARPC.
ARC	16H	Application Response Code used for ARPC generation followed by 12 zeros (ARC || 00 00 00 00 00 00). ***Only if Mode is 1 or 2.***

Response Body:

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Data	Length & Type	Description
Command Identifier	4H	**0503**.
Error Code	8H	Error code received from the HSM: **00000000** - No error code. **xxxxxxxx** - Other code defined in Error Codes.
ARQC Validation Result	1N	The validation result: **0** - Correct. **1** - Failed. ***Only if Mode is 0 or 1.***
ARPC	4H	Calculated ARPC. ***Only if Mode is 2 or 1 and no error is encountered.***
Diagnostic Data	16H	Calculated ARQC/TC/AAC value. It is returned if **ARQC Validation Result** is **1** and the HSM is in Authorised State. ***Only if Mode is 0 or Mode is 1.***

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3.2.4 0508 - ARQC Verification and/or ARPC Generation (EMV 4.x)

To validate an ARQC (or TC/AAC) and, optionally, to generate an ARPC. Alternatively, the command can be used to generate an ARPC alone. This function is a general purpose command which will validate an ARQC, TC or AAC.

Diagnostic data is produced by this command if the HSM is in Authorized State. The diagnostic data consists of a generated ARQC, which is returned to the host if verification of the supplied ARQC fails.

This command supports the following cryptogram methods:

Application	CVN	Scheme
VISA VIS	14	0
	18 or 22	2 or 3
MasterCard M/Chip	12 or 13	0
	14 or 15	2

It is the responsibility of the host system to add any scheme specific padding data to the end of the transaction data. For some schemes this means appending a byte containing hex 80 to the end of the data. This command adds no further padding data in any case.

Request Body:

Data	Length & Type	Description
Command Identifier	4H	**0508**.
Mode	1N	Execution mode: **0** - Validate ARQC. **1** - Validate ARQC and calculate ARPC EMV 4.1 method 1. **2** - Calculate ARPC EMV 4.1 method 1. **3** - Validate ARQC and calculate ARPC EMV 4.1 method 2. **4** - Calculate ARPC EMV 4.1 method 2.
Scheme	1N	Session Key operation: **0** - Method A ICC MK Derivation and EMV2000 SKD. **1** - Method B ICC MK Derivation and EMV2000 SKD. **2** - Method A ICC MK Derivation and EMV Common SKD. **3** - Method B ICC MK Derivation and EMV Common SKD. **4** - M/CHIP4 using Mastercard Proprietary SKD.
MK-AC	varK	Key for Application Cryptograms (under LMK29 or LMK6).
IV-AC	32H	IV for EMV2000 session Key Derivation. ***Only if Scheme is 0 or 1.***
PAN/PSN Length	2N	Pre-formatted PAN/PAN Sequence Number length (**16**..**20**). ***Only if Scheme is 1 or 3.***

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Data	Length & Type	Description
PAN/PSN	16N or varN	For Scheme = 0, 2 or 4, this field will be fixed at 16 bytes, and should contain the pre-formatted PAN/PAN Sequence Number. For Scheme ID = 1 or 3, the field length is specified by the **PAN/-PANseqNo Length** field. It is the responsibility of the host system to ensure that the PAN/-PAN Sequence Number is appropriately padded.
Branch/Height Parameters	1N	Data for EMV2000 Session Key Derivation: **0** - Branch factor 2; Tree Height 16. **1** - Branch factor 4; Tree Height 8. ***Only if Scheme is 0 or 1.***
ATC	4H	The Application Transaction Counter from the card. This is used for session key generation.
UN	8H	The Unpredictable Number. This is used for session key generation. ***Only if Scheme is 4.***
Transaction Data Length	3N	Next field length. Can be any length from 1 to 255 bytes. ***Only if Mode is 0, 1 or 3.***
Transaction Data	varH	If the data supplied is a multiple of 8 bytes, no extra padding is added. If it is not a multiple of 8 bytes, additional zero padding is added. Note: If alternative padding methods are required, it is the responsibility of the host to provide this. ***Only if Mode is 0, 1 or 3.***
ARQC/TC/AAC	16H	ARQC/TC/AAC to validate and/or use to generate ARPC.
ARC	4H	Application Response Code used for ARPC generation. ***Only if Mode is 1 or 2.***
CSU	8H	Card Status Update used for ARPC generation for Common Core Definitions (CCD) cards. ***Only if Mode is 3 or 4.***
Proprietary Authentication Data Length	2N	Next field length. Can be any length from 1 to 16 bytes. ***Only if Mode is 3 or 4.***
Proprietary Authentication Data	varH	Contains optional issuer data for transmission to the card in the Issuer Authentication Data of an online transaction. ***Only if Mode is 3 or 4.***

Response Body:

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Data	Length & Type	Description
Command Identifier	4H	**0508**.
Error Code	8H	Error code received from the HSM: **00000000** - No error code. **xxxxxxxx** - Other code defined in Error Codes.
ARQC Validation Result	1N	**0** - Correct. **1** - Failed. ***Only if Mode is 0, 1 or 3.***
ARPC	16H	Calculated ARPC value. It is returned if **Validation Result** is **0**. ***Only if Mode is 1, 2, 3 or 4 and no error is encountered.***
Diagnostic Data	16H	Calculated ARQC/TC/AAC value. It is returned if **Validation Result** is **1** and the HSM is in Authorised State. ***Only if Mode is 0, 1 or 3.***

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3.2.5 0509 - Generate Secure Message (EMV 3.1.1)

Generate a Secure Message with Integrity over data to be sent from the Issuer back to the Card. Optionally, Secure Messaging with Confidentiality is provided in which case the message data shall be supplied encrypted under a Transport Key. In this latter case the command first decrypts the message data using the Transport Key before re-encrypting it using a Session Key. This command is also used to change or unblock a PIN.

To change the PIN held by an EMV card, the issuer has to validate the existing PIN, and then accept a new PIN in a standard PIN block format. This PIN block is then translated from a standard ATM PIN block format (encrypted under a terminal key) to an application specific PIN block format (encrypted under a confidentiality session key).

To generate a PIN unblock script, use **0** in the **Mode** field (integrity only), with an EMV PIN Unblock APDU supplied in the **Plaintext Message Data** field.

This command supports:

Application	CVN
VISA VIS	10 or 18
MasterCard M/Chip	10 or 11

Request Body:

Data	Length & Type	Description
Command Identifier	4H	**0509**.
Mode	1N	Operation to perform: **0** - Provide only integrity functionality. **1** - Provide integrity and confidentiality using the same Issuer Master Key. **2** - Provide integrity and confidentiality using different Issuer Master Keys. **3** - Provide integrity and PIN block translation for PIN change, using the same Issuer Master Key. **4** - Provide integrity and PIN block translation for PIN change, using different Issuer Master Keys.
Scheme	1N	Product: **0** - Visa. **1** - Mastercard. **2** - American Express.
MK-SMI	varK	The Master Key used to generate a MAC on the message (under LMK31 or LMK6).
PAN	16N	16 right digits of the PAN. If PAN length is less than 16, pad with zeros to the left.

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Data	Length & Type	Description
PSN	2N	The PAN Sequence Number.
Integrity Session Data	16H	Data for Integrity Session Key derivation: ATC || 00 00 00 00 00 00 for Visa. RANDi Data for MasterCard.
Plaintext Message Length	4N	Next field length.
Plaintext Message	varH	Plaintext message data.
MK-SMC	varK	The Master Key used to encrypt the message (under LMK30 or LMK6). ***Only if Mode is 2 or 4.***
TK	varK	The Trasnport Key used to decrypt the supplied message (under LMK7). ***Only if Mode is 1 or 2.***
Confidentiality Session Data	16H	Data for confidentiality Session Key derivation: ATC || 00 00 00 00 00 00 for Visa. RANDc Data for MasterCard. ***Only if Mode is NOT 0.***
Offset	4N	Position within Plaintext data to insert Ciphertext data. Shall be between '0000' and Plaintext Message Data length. If Offset = x, Ciphertext is inserted AFTER the xth byte of the Plaintext (i.e. if length of Plaintext data is '0062', and Offset is '0062', Ciphertext data is placed at the end of the plaintext message). ***Only if Mode is 1 or 2.***
Cipher Text Message Data Length	4N	Next field length. Shall be multiple of 16 bytes. ***Only if Scheme is 1 or 2.***
Cipher Text Message Data	varH	Cipher Text Message supplied encrypted using a Transport Key (TK). Note that no additional padding is performed on the decrypted message before the re-encryption process. ***Only if Scheme is 1 or 2.***
Source PIN Encryption Key	varK	The ATM encryption PIN Block Key, used to decrypt the supplied PIN block (under LMK3). ***Only if Mode is 3 or 4.***

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Data	Length & Type	Description
Input PIN Block Format	1N	Input PIN block format: **0** - ISO0. **1** - ISO1. **2** - ISO2. **3** - ISO3. **4** - IBM3624. **5** - Diebold. ***Only if Mode is 3 or 4.***
Output PIN Block Format	1N	Output PIN block format: **0** - PIN Block I: Standard EMV PIN block. **1** - PIN Block II: Visa format without using current PIN. **2** - PIN Block III: Visa format using current PIN. **3** - PIN Block IV: MasterCard Pay Now & Pay Later. ***Only if Mode is 3 or 4.***
Current PIN Block	16H	Current PIN block received from ATM. ***Only present if Mode is 3 or 4 and Output PIN Block Format is 2.***
New PIN Block	16H	New PIN block received from ATM. ***Only present if Mode is 3 or 4.***
MK-AC Key	varK	MK-AC Key (under LMK29 or LMK6). The Issuer Master Key for generating and verifying Application Cryptograms. This is required to create PIN Blocks for Visa PIN Change. ***Only present if Mode is 3 or 4 AND Output PIN Block Format is 1 or 2.***

Response Body:

Data	Length & Type	Description
Command Identifier	4H	**0509**.
Error Code	8H	Error code received from the HSM: **00000000** - No error code. **xxxxxxxx** - Other code defined in Error Codes.
Script Signature	16H	MAC perform over the script command.
Re-encrypted ciphertext Data Length	6N	Next field length. ***Only if Mode is 1, 2, 3 or 4.***
Re-encrypted ciphertext Data	varH	Re-encrypted ciphertext message data. ***Only if Mode is 1, 2, 3 or 4.***

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Data	Length & Type	Description

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3.2.6 0510 - Generate Secure Message (EMV 4.x)

Generate a Secure Message with Integrity over data to be sent from the Issuer back to the Card. Optionally, Secure Messaging with Confidentiality is provided in which case the message data shall be supplied encrypted under a Transport Key. In this latter case the command first decrypts the message data using the Transport Key before re-encrypting it using a Session Key.

This command performs a similar function to the KU command, but uses the EMV2000 method or EMV Common Session Key Derivation method for generating the session key.

This command supports:

Application	CVN	Scheme
VISA VIS	14	0
MasterCard M/Chip	12 or 13	1
	14 or 15	4

Request Body:

Data	Length & Type	Description
Command Identifier	4H	**0510**.
Mode	1N	Product: **0** - Provide integrity. **1** - Provide integrity and confidentiality. **2** - Provide integrity and PIN change.
Scheme	1N	Session Key Derivation Method: **0** - Visa VIS using EMV 4.x Option A ICC Master Key Derivation and EMV2000 Session Key Derivation. **1** - MasterCard M/Chip using EMV 4.x Option A ICC Master Key Derivation and EMV2000 Session Key Derivation. **2** - CCD using EMV 4.x Option B ICC Master Key Derivation and EMV2000 Session Key Derivation. **3** - Visa VIS using EMV 4.x Option A ICC Master Key Derivation and EMV Common Session Key Derivation. **4** - MasterCard M/Chip using EMV 4.x Option A ICC Master Key Derivation and EMV Common Session Key Derivation. **5** - CCD using EMV 4.x Option B ICC Master Key Derivation and EMV Common SKD.
SMI-MK	varK	The Master Key used to generate a MAC on the message (under LMK31 or LMK6).

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Data	Length & Type	Description
SMI-IV	32H	IV for EMV2000 Secure Messaging for Integrity session key derivation. ***Only if Scheme is 0, 1 or 2.***
PAN/PSN Length	2N	Number of bytes of the PAN/PSN field. ***Only if Scheme is 2 or 5.***
PAN/PSN	16/varN	If **Scheme** is 0, 1, 3 or 4 this field shall be the 16 rigth most digits of the PAN and PAN Sequence Number.[1] If **Scheme** is 2 or 5, this field contains the whole PAN and the PAN Sequence Number with a length specified by the PAN Length field. [2,3]
Branch/Height Parameters	1N	Data for EMV2000 Session Key Derivation: **0** - Branch factor 2; Tree Height 16. **1** - Branch factor 4; Tree Height 8. ***Only if Scheme is 0, 1 or 2.***
ATC	4H	The Application Transaction Counter from the card. This is used for session key generation. ***Only if Scheme is 0, 1 or 2.***
Application Cryptogram	16H	The Application Cryptogram returned by the card in response to the first GENERATE AC command. ***Only if Scheme is 3, 4 or 5.***
Plaintext Message Length	4N	Next field length.
Plaintext Message	varH	Plaintext Message Data.
SMC-MK	varK	The Master Key used to encrypt the message (under LMK30 or LMK6). ***Only if Mode is 1 or 2.***
SMC-IV	32H	IV for EMV2000 Secure Messaging for confidentiality session key derivation. ***Only if Mode is 1 or 2 AND Scheme is 0, 1 or 2.***
TK	varK	The Transport key used to decrypt the supplied message (under LMK7). ***Only if Mode is 1.***
Offset	4N	Position within Plaintext data to insert Ciphertext data. Shall be between ‘0000’ and Plaintext Message Data length. If Offset = x, Ciphertext is inserted AFTER the xth byte of the Plaintext (i.e. if length of Plaintext data is ‘0062’, and Offset is ‘0062’, Ciphertext data is placed at the end of the plaintext message). ***Only if Mode is 1.***
Cipher Text Message Data Length	4N	Next field length. Shall be multiple of 16 bytes. ***Only if Mode is 1.***

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Data	Length & Type	Description
Cipher Text Message Data	varH	Cipher Text Message supplied encrypted using a Transport Key (TK). It shall be multiple of 16 bytes long. Note that no additional padding is performed on the decrypted message before the re-encryption process. ***Only if Mode is 1.***
Source PIN Encryption Key	varK	The ATM encryption PIN Block Key, used to decrypt the supplied PIN block (under LMK3). ***Only if Mode is 2.***
Input PIN Block Format	1N	Input PIN block format: **0** - ISO0. **1** - ISO1. **2** - ISO2. **3** - ISO3. **4** - IBM3624. **5** - Diebold. ***Only if Mode is 2.***
Output PIN Block Format	1N	Output PIN block format: **0** - PIN Block I: Standard EMV PIN block. **1** - PIN Block II: Visa format without using current PIN. **2** - PIN Block III: Visa format using current PIN. **3** - PIN Block IV: MasterCard Pay Now & Pay Later. ***Only if Mode is 2.***
Current PIN Block	16H	Current PIN block received from ATM. ***Only if Mode is 2 and Output PIN Block Format is 2.***
New PIN Block	16H	New PIN block received from ATM. ***Only if Mode is 2.***
MK-AC Key	varK	MK-AC Key (under LMK29 or LMK6). The Issuer Master Key for generating and verifying Application Cryptograms. This is required to create PIN Blocks for Visa PIN Change. ***Only if Mode is 2 AND Output PIN Block Format is 1 or 2.***

[1] It is the responsibility of the host system to ensure that the PAN/PSN is appropriately padded to 16 characters (8 bytes), according to EMV Option A.

[2] It is the responsibility of the host system to ensure that the PAN/PSN is an even number of characters (therefore a whole number of bytes) by pre-pending an extra 0 digit to the left if necessary. The PAN/PSN is then used as in EMV Option B.

[3] If PAN/PSN Length is less than 16 characters (8 bytes) EMV Option A will be used, and the HSM will pad appropriately to 16 chars if necessary.

Response Body:

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Data	Length & Type	Description
Command Identifier	4H	**0510**.
Error Code	8H	Error code received from the HSM: **00000000** - No error code. **xxxxxxxx** - Other code defined in Error Codes.
Script Signature	16H	MAC perform over the script command.
Re-encrypted Ciphertext Data Length	6N	Next field length. ***Only if Mode is 1 or 2.***
Re-encrypted Ciphertext Data	varH	Re-encrypted ciphertext message data. ***Only if Mode is 1 or 2.***

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3.2.7 0511 - Mastercard IAV Validation

Verify a Mastercard IAV.

Request Body:

Data	Length & Type	Description
Command Identifier	4H	**0511**.
Key	varK	IAV key (under LMK43 or LMK8).
Hash Algorithm	2N	The hash algorithm to be used on the HMAC: **11** - SHA-224. **12** - SHA-256. **13** - SHA-384. **14** - SHA-512. **20** - SHA-512/224. **21** - SHA-512/256.
AAV Length	2N	Next field length.
AAV	varH	Base64-encoded Accountholder Authentication Value. 21 bytes are expected once decoded.
RFU	4H	Reserved for Future Use. ***Value: C000.***
PAN Length	2N	Next field length. Up to 20 digits.
PAN	varN	The PAN represented as binary-coded decimal. Will be left justified and padded on the right with hexadecimal 'F's to a length of 20 digits.
DS Transaction ID	32H	Universally unique transaction identifier assigned by the Directory Server to identify a single transaction. Binary format.

Response Body:

Data	Length & Type	Description
Command Identifier	4H	**0511**.
Error Code	8H	Error code received from the HSM: **00000000** - No error code. **xxxxxxxx** - Other code defined in Error Codes.
Validation Result	1N	Validation result: **0** - Correct. **1** - Failed.

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3.3 PIN Calculation (Cryptosec-Banking Only)

The Cryptosec-Banking HSM provides commands to support the PIN/Offset generation.

These commands are capable of rejecting 'weak' PINs (i.e. PINs that are considered easily guessed by someone other than the legitimate cardholder) defined by passing them into the HSM in a 'Weak PIN List' host command.

3.3.1 0605 - PIN Export

Generate a PIN Block using the provided offset.

Request Body:

Data	Length & Type	Description
Command Identifier	4H	**0605**.
PIN Key	varK	PIN Calculation Key (under LMK46 or LMK4).
PIN Encryption Key	varK	PIN block transport key (under LMK3).
DT	16/var[1] N/H/K[2]	Decimalization table.[3]
PIN Block Format	1N	Output PIN block format: **0** - ISO0.[4] **3** - ISO3.[4] **7** - ISO4.[5]
PAN Length	2N	Next field length.
PAN	varN	The PAN of the card.
Offset Length	2N	Next field length.
Offset	varN	The offset.
Random Number	10H	Random number to build the PIN block.+ ***Only if PIN Block Format is 3.***

[1] Decimalization table length can be:
16 for a raw table.
var for a table in a container version 1.
[2] Decimalization table type can be:
N for a raw table in the clear.
H for an encrypted raw table.
K for a table in a container version 1, encrypted or in the clear.
[3] PCI PTS HSM v4.0-Approved mode of operation requires an encrypted raw table or a table in a container version 1. For new implementations, container version 1 is recommended.
[4] Only if **PIN Encryption Key** is a DES key.
[5] Only if **PIN Encryption Key** is an AES key.

Response Body:

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Data	Length & Type	Description
Command Identifier	4H	**0605**.
Error Code	8H	Error code received from the HSM: **00000000** - No error code. **xxxxxxxx** - Other code defined in Error Codes.
Encrypted PIN Block	16/32H[1]	The PIN block.

[1] 32 if **PIN Encryption Key** is an AES key, 16 otherwise.

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3.3.2 0608 - PVV Calculation

Generate a PIN from interchange using the VISA PVV method.

Request Body:

Data	Length & Type	Description
Command Identifier	4H	**0608**.
PIN Key	varK	PIN Calculation Key (under LMK47 or LMK4).
PIN Encryption Key	varK	PIN Block transport key (under LMK3).
PIN Block Format	1N	PIN block format: **0** - ISO0.[1] **1** - ISO1.[1] **2** - ISO2.[1] **3** - ISO3.[1] **4** - IBM3624.[1] **5** - Diebold.[1] **6** - Realsec1.[1] **7** - ISO4.[2]
PIN Block	16/32H[3]	The encrypted PIN block.
PAN Length	2N	Next field length.
PAN	varN	The PAN of the card.
Padding Value	2H	The padding value used to create the PIN block. ***Only if PIN Block Format is 4.***
Key Index	1N	Key index digit: **1..6**.

[1] Only if **Source Encryption Key** is a DES key.
[2] Only if **Source Encryption Key** is an AES key.
[3] 32 if **Source Encryption Key** is an AES key, 16 otherwise.

Response Body:

Data	Length & Type	Description
Command Identifier	4H	**0608**.
Error Code	8H	Error code received from the HSM: **00000000** - No error code. **xxxxxxxx** - Other code defined in Error Codes.
PVV	4N	PVV value.

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3.3.3 0611 - PIN Block Calculation from a PIN

Generate a random PIN or get a clear PIN and return it as an encrypted PIN Block.

NOTICE: This command is not available if the HSM is configured in PCI PTS HSM v4.0-Approved mode.

Request Body:

Data	Length & Type	Description
Command Identifier	4H	**0611**.
PIN Encryption Key	varK	PIN Block transport key (under LMK3).
PIN Block Format	1N	PIN block format: **0** - ISO0.[1] **1** - ISO1.[1] **2** - ISO2.[1] **3** - ISO3.[1] **4** - IBM3624.[1] **5** - Diebold.[1] **6** - Realsec1.[1] **7** - ISO4.[2]
PAN Length	2N	Next field length.
PAN	varN	The PAN of the card.
Padding Value	1H	The padding value use to create the PIN block. ***Only if PIN Block Format is 4.***
Mode	1N	PIN generation mode: **0** - Provided clear PIN. **1** - Internally generated random PIN.
PIN Length	1N	The length of the PIN. ***Next field length if Mode is 0.*** ***The length of the PIN to generate if Mode is 1.***
PIN	varN	The clear PIN. ***Only if Mode is 0.***

[1] Only if **PIN Encryption Key** is a DES key.
[2] Only if **PIN Encryption Key** is an AES key.

Response Body:

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Data	Length & Type	Description
Command Identifier	4H	**0611**.
Error Code	8H	Error code received from the HSM: **00000000** - No error code. **xxxxxxxx** - Other code defined in Error Codes.
Encrypted PIN Block	16/32H[1]	The PIN block.

[1] 32 if **PIN Encryption Key** is an AES key, 16 otherwise.

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3.3.4 0618 - PIN Offset Generation

Generate a PIN offset using the selected algorithm.

Request Body:

Data	Length & Type	Description
Command Identifier	4H	**0618**.
PIN Key	varK	PIN calculation key (under LMK45 (Netherlands PIN 1, or IBM German Bank Pool Institution), LMK46 (IBM3624 PIN Offset) or LMK4).
PIN Encryption Key	varK	PIN block transport key (under LMK3).
DT	16/var[1] N/H/K[2]	Decimalization table.[3]
PIN Block Format	1N	Output PIN block format: **0** - ISO0.[4] **1** - ISO1.[4] **2** - ISO2.[4] **3** - ISO3.[4] **4** - IBM3624.[4] **5** - Diebold.[4] **6** - Realsec1.[4] **7** - ISO4.[5]
Algorithm	1N	PIN generation algorithm: **1** - IBM3624 PIN Offset. **2** - Netherlands PIN 1. **4** - IBM German Bank Pool Institution.
PIN Block	16/32H[6]	Encrypted PIN block.
PAN Length	2N	Next field length.
PAN	varN	The PAN of the card.
Padding Value	2H	The padding value use to create the PIN block. ***Only if PIN Block Format is 4.***
PIN Length	1N	Number of digits of the PIN. ***Only if Algorithm is 1.***
Validation Data	16H	The validation data for PIN Generation. ***Only if Algorithm is 1 or 4.***

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Data	Length & Type	Description

[1] Decimalization table length can be:
16 for a raw table.
var for a table in a container version 1.
[2] Decimalization table type can be:
N for a raw table in the clear.
H for an encrypted raw table.
K for a table in a container version 1, encrypted or in the clear.
[3] PCI PTS HSM v4.0-Approved mode of operation requires an encrypted raw table or a table in a container version 1. For new implementations, container version 1 is recommended.
[4] Only if **PIN Encryption Key** is a DES key.
[5] Only if **PIN Encryption Key** is an AES key.
[6] 32 if **PIN Encryption Key** is an AES key, 16 otherwise.

Response Body:

Data	Length & Type	Description
Command Identifier	4H	**0618**.
Error Code	8H	Error code received from the HSM: **00000000** - No error code. **xxxxxxxx** - Other code defined in Error Codes.
Offset	varN	Returned offset value.

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3.3.5 0619 - PIN Generation

Calculate a PIN according to a generation algorithm and return it as an encrypted PIN Block.

Request Body:

Data	Length & Type	Description
Command Identifier	4H	**0619**.
PIN Key	varK	PIN calculation key (under LMK45 (IBM German Bank Pool Institution or Interbank), LMK46 (IBM3624 or IBM3624 PIN Offset) or LMK4).
PIN Encryption Key	varK	PIN block transport key (under LMK3).
DT	16/var[1] N/H/K[2]	Decimalization table.[3]
PIN Block Format	1N	Output PIN block format: **0** - ISO0.[4] **1** - ISO1.[4] **2** - ISO2.[4] **3** - ISO3.[4] **4** - IBM3624.[4] **5** - Diebold.[4] **6** - Realsec1.[4] **7** - ISO4.[5]
PIN Generation Algorithm	1N	PIN generation algorithm: **0** - IBM3624. **1** - IBM3624 PIN Offset. **4** - IBM German Bank Pool Institution. **5** - Interbank.
PAN Length	2N	Next field length.
PAN	varN	The PAN of the card.
Padding Value	1H	Padding value. ***Only if PIN Block Format is 4.***
PIN Length	1N	PIN length. ***Only if PIN Generation Algorithm is 0.***
Key Indicator	1H	Key indicator. ***Only if PIN Generation Algorithm is 5.***
Validation Field	3H	Validation field. ***Only if PIN Generation Algorithm is 5.***
PIN Offset Length	1N	Next field length. ***Only if PIN Generation Algorithm is 1.***
PIN Offset	varH	The PIN offset. ***Only if PIN Generation Algorithm is 1.***

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Data	Length & Type	Description
Validation Data	16H	The validation data for PIN Generation.

[1] Decimalization table length can be:
16 for a raw table.
var for a table in a container version 1.
[2] Decimalization table type can be:
N for a raw table in the clear.
H for an encrypted raw table.
K for a table in a container version 1, encrypted or in the clear.
[3] PCI PTS HSM v4.0-Approved mode of operation requires an encrypted raw table or a table in a container version 1. For new implementations, container version 1 is recommended.
[4] Only if **PIN Encryption Key** is a DES key.
[5] Only if **PIN Encryption Key** is an AES key.

Response Body:

Data	Length & Type	Description
Command Identifier	4H	**0619**.
Error Code	8H	Error code received from the HSM: **00000000** - No error code. **xxxxxxxx** - Other code defined in Error Codes.
Encrypted PIN Block	16/32H[1]	The PIN block.

[1] 32 if **PIN Encryption Key** is an AES key, 16 otherwise.

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3.4 PIN Verification (Cryptosec-Banking Only)

The Cryptosec-Banking HSM provides support for many PIN verification techniques. The techniques often involve the generation of a PIN using a given algorithm, or an algorithm that requires a value known as an offset.

3.4.1 0602 - PIN Verification

Verify a PIN provided in an encrypted PIN Block.

Request Body:

Data	Length & Type	Description
Command Identifier	4H	**0602**.
PIN Calculation Key	varK	PIN calculation key (under LMK45 (IBM German Bank Pool Institution, Interbank, ALT_1 PIN Offset or H 24), LMK46 (IBM3624 or IBM3624 PIN Offset), LMK47 (VISA PVV) or LMK4).
PIN Encryption Key	varK	PIN block transport key (under LMK3).
DT	16/var[1] N/H/K[2]	Decimalization table.[3]
PIN Block Format	1N	PIN block format: **0** - ISO0.[4] **1** - ISO1.[4] **2** - ISO2.[4] **3** - ISO3.[4] **4** - IBM3624.[4] **5** - Diebold.[4] **6** - Realsec1.[4] **7** - ISO4.[5]
Algorithm	1N	PIN generation algorithm: **0** - IBM3624. **1** - IBM3624 PIN Offset. **4** - IBM German Bank Pool Institution. **5** - Interbank. **6** - ALT_1 PIN Offset. **7** - H24
Legacy Field	1N	Mandatory but not used: **0** - Not used.
PIN Block	16/32H[6]	The PIN block to be verified.

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Data	Length & Type	Description
PIN Extraction Method	2N	PIN extraction method: **00** - Do not use any padding value. **01** - Use digit specified in **Padding Value** field. **02** - First hexadecimal digit found in the decrypted PIN Block. **03** - Last digit found in the decrypted PIN Block. **04..16** - PIN length indicated by this value. ***Only if PIN Block Format is 4.***
PAN Length	2N	Next field length.
PAN	varN	The PAN of the card.
Padding Value	1H	The padding value use to create the PIN block. ***Only if PIN Block Format is 4 and PIN Extraction Method is 01.***
PIN Length	1N	Number of digits of the PIN. ***Only if Algorithm is 0.***
PIN Offset Length	1N	Next field length. ***Only if PIN Generation Algorithm is 1 or 6***
PIN Offset	varH	The PIN offset. ***Only if PIN Generation Algorithm is 1 or 6.***
Key Indicator	1H	Key indicator. ***Only if PIN Generation Algorithm is 5.***
Validation Field	3H	Validation field. ***Only if PIN Generation Algorithm is 5.***
Extraction PAN Length	1N	Next field length. ***Only if PIN Block Format is NOR 4 neither 5 and it is different to the PAN field***
Extraction PAN	varH	The PAN used to extract the PIN from the **PIN Block**. ***Only if PIN Block Format is NOR 4 neither 5 and it is different to the PAN field.***

[1] Decimalization table length can be:
16 for a raw table.
var for a table in a container version 1.
[2] Decimalization table type can be:
N for a raw table in the clear.
H for an encrypted raw table.
K for a table in a container version 1, encrypted or in the clear.
[3] PCI PTS HSM v4.0-Approved mode of operation requires an encrypted raw table or a table in a container version 1. For new implementations, container version 1 is recommended.
[4] Only if **Source Encryption Key** is a DES key.
[5] Only if **Source Encryption Key** is an AES key.
[6] 32 if **Source Encryption Key** is an AES key, 16 otherwise.

Response Body:

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Data	Length & Type	Description
Command Identifier	4H	**0602**.
Error Code	8H	Error code received from the HSM: **00000000** - No error code. **xxxxxxxx** - Other code defined in Error Codes.
Verification Result	1N	PIN verification result: **0** - Correct. **1** - Failed.

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3.4.2 0607 - PVV Verification

Verify a PIN using the VISA PVV method.

Request Body:

Data	Length & Type	Description
Command Identifier	4H	**0607**.
PIN Key	varK	PIN calculation key (under LMK4).
PIN Encryption Key	varK	PIN Block transport key (under LMK3).
PIN Block Format	1N	PIN block format: **0** - ISO0.[1] **1** - ISO1.[1] **2** - ISO2.[1] **3** - ISO3.[1] **4** - IBM3624.[1] **5** - Diebold.[1] **6** - Realsec1.[1] **7** - ISO4.[2]
PIN Block	16/32H[3]	The encrypted PIN block.
PIN Extraction Method	2N	PIN extraction method: **00** - Do not use any padding value. **01** - Use digit specified in **Padding Value** field. **02** - First hexadecimal digit found in the decrypted PIN Block. **03** - Last digit found in the decrypted PIN Block. **04..16** - PIN length indicated by this value. ***Only if PIN Block Format is 4.***
PAN Length	2N	Next field length.
PAN	varN	The PAN of the card.
Padding Value	1H	The padding value used to create the PIN block. ***Only if PIN Block Format is 4 and PIN Extraction Method is 00 or 01.***
Key Index	1N	Key index digit: **1..6**.
PVV	4H	The PVV to verity.

[1] Only if **Source Encryption Key** is a DES key.
[2] Only if **Source Encryption Key** is an AES key.
[3] 32 if **Source Encryption Key** is an AES key, 16 otherwise.

Response Body:

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Data	Length & Type	Description
Command Identifier	4H	**0607**.
Error Code	8H	Error code received from the HSM: **00000000** - No error code. **xxxxxxxx** - Other code defined in Error Codes.
Verification Result	1N	PVV verification result: **0** - Correct. **1** - Failed.

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3.4.3 0612 - Load weak PINs List

Load a "Weak PIN list" into the HSM, so any time the HSM generates a random PIN it will be checked against that list to ensure it does not match one of the entries. Load a "Weak PIN list" into the HSM, so any time the HSM generates a random PIN it will be checked against that list to ensure it does not match one of the entries

Request Body:

Data	Length & Type	Description
Command Identifier	4H	**0612**.
PIN Length	2N	Number of digits of each weak PIN to be informed.
Number of Weak PINs	4N	Number of PINs to be informed.
Weak PIN List	varN	Weak PINs list to load into the HSM.

Response Body:

Data	Length & Type	Description
Command Identifier	4H	**0612**.
Error Code	8H	Error code received from the HSM: **00000000** - No error code. **xxxxxxxx** - Other code defined in Error Codes.

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3.4.4 0615 - PVV Verification DUKPT

Verify a PIN from interchange using the VISA PVV method.

For DES Base Derivation Keys, the ANSI X9.24-1:2009 method for DUKPT key derivation is used. For AES Base Derivation Keys, the X9.24-3:2017 method for DUKPT key derivation is used.

Request Body:

Data	Length & Type	Description
Command Identifier	4H	**0615**.
PIN Key	varK	The PIN calculation key (under LMK4).
Base Derivation Key	varK	The Base derivation key (under LMK19).
Delimiter	1A	**;** (0x3B ASCII) for TDES-DUKPT.[1] ***Optional. If present, KSN Descriptor field shall exist.***
KSN Descriptor	3N	The descriptor for the **KSN** for TDES-DUKPT.[1]
KSN	12-20H	Current Key Serial Number for TDES-DUKPT.[1] ***If the KSN Descriptor is not present, this value shall be 20 digits in length.***
Working Key Type	2N	The type of the key to derive for AES-DUKPT:[2] **02** - AES128. **03** - AES192. **04** - AES256.
Initial Key ID	16H	The Initial Key ID for AES-DUKPT.[2]
Transaction Counter	8H	The Transaction Counter for AES-DUKPT.[2]
PIN Block Format	1N	PIN block format: **0** - ISO0.[1] **1** - ISO1.[1] **2** - ISO2.[1] **3** - ISO3.[1] **4** - IBM3624.[1] **5** - Diebold.[1] **6** - Realsec1.[1] **7** - ISO4.[2]
PIN Block	16/32H[3]	The encrypted PIN block.
PIN Extraction Method	2N	PIN extraction method: **00** - Do not use any padding value. **01** - Use digit specified in **Padding Value** field. **02** - First hexadecimal digit found in the decrypted PIN Block. **03** - Last digit found in the decrypted PIN Block. **04..16** - PIN length indicated by this value. ***Only if PIN Block Format is 4.***

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Data	Length & Type	Description
PAN Length	2N	Next field length.
PAN	varN	The PAN of the card.
Padding Value	1H	The padding value use to create the PIN block. ***Only if PIN Block Format is 4 and PIN Extraction Method is 00 or 01.***
Key Index	1N	Key index digit.
PVV	4H	The PVV to verity.

[1] Only if **Base Derivation Key** is a DES key.
[2] Only if **Base Derivation Key** is an AES key.
[3] 32 if **Base Derivation Key** is an AES key, 16 otherwise.

Response Body:

Data	Length & Type	Description
Command Identifier	4H	**0615**.
Error Code	8H	Error code received from the HSM: **00000000** - No error code. **xxxxxxxx** - Other code defined in Error Codes.
Verification Result	1N	PVV verification result: **0** - Correct. **1** - Failed.

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3.4.5 0616 - PIN Verification DUKPT

Verify a PIN.

The following derivation methods can be used:

- ANSI X9.24-1:2009 method for TDES-DUKPT key derivation.
- ANSI X9.24-3-2017 method for AES-DUKPT key derivation.

Request Body:

Data	Length & Type	Description
Command Identifier	4H	**0616**.
PIN Key	varK	The PIN calculation key (under LMK45 (IBM German Bank Pool Institution, Interbank, ALT_1 PIN Offset or H 24), LMK46 (IBM3624 or IBM3624 PIN Offset), LMK47 (VISA PVV) or LMK4).
Base Derivation Key	varK	The base derivation key (under LMK19).
DT	16/var[1] N/H/K[2]	Decimalization table.[3]
PIN Block Format	1N	PIN block format: **0** - ISO0.[4] **1** - ISO1.[4] **2** - ISO2.[4] **3** - ISO3.[4] **4** - IBM3624.[4] **5** - Diebold.[4] **6** - Realsec1.[4] **7** - ISO4.[5]
Algorithm	1N	PIN generation algorithm: **0** - IBM3624. **1** - IBM3624 PIN Offset. **4** - IBM German Bank Pool Institution. **5** - Interbank. **6** - ALT_1 PIN Offset. **7** - H24
DUKPT Mode	1N	Derivation method: **1** - DES-DUKPT (deprecated). **2** - 3DES-DUKPT or AES-DUKPT.
PIN Block	16/32H[6]	The encrypted PIN block.

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Data	Length & Type	Description
Delimiter	1A	**;** (0x3B ASCII) for TDES-DUKPT.[4] ***Optional. If present, KSN Descriptor field shall exist.***
KSN Descriptor	3N	The descriptor for the **KSN** for TDES-DUKPT.[4]
KSN	12-20H	Current Key Serial Number for TDES-DUKPT.[4] ***If the KSN Descriptor is not present, this value shall be 20 digits in length.***
Working Key Type	2N	The type of the key to derive for AES-DUKPT.[5] **02** - AES128. **03** - AES192. **04** - AES256.
Initial Key ID	16H	The Initial Key ID for AES-DUKPT.[5]
Transaction Counter	8H	The Transaction Counter for AES-DUKPT.[5]
PIN Extraction Method	2N	PIN extraction method: **00** - Do not use any padding value. **01** - Use digit specified in **Padding Value** field. **02** - First hexadecimal digit found in the decrypted PIN Block. **03** - Last digit found in the decrypted PIN Block. **04..16** - PIN length indicated by this value. ***Only if PIN Block Format is 4.***
PAN Length	2N	Next field length.
PAN	varN	The PAN of the card.
Padding Value	1H	The padding value use to create the PIN block. ***Only if PIN Block Format is 4 and PIN Extraction Method is 00 or 01.***
PIN Length	1N	Number of digits of the PIN. ***Only if Algorithm is 0.***
Offset Length	1N	Next field length. ***Only if Algorithm is 1, 3, 6 or 7.***
Offset	varN	The offset. ***Only if Algorithm is 1, 3, 6 or 7.***
Key Indicator	1N	Key indicator digit. ***Only if Algorithm is 5.***
Validation Field	3H	Validation data field. ***Only if Algorithm is 5.***

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Data	Length & Type	Description

[1] Decimalization table length can be:
16 for a raw table.
var for a table in a container version 1.
[2] Decimalization table type can be:
N for a raw table in the clear.
H for an encrypted raw table.
K for a table in a container version 1, encrypted or in the clear.
[3] PCI PTS HSM v4.0-Approved mode of operation requires an encrypted raw table or a table in a container version 1. For new implementations, container version 1 is recommended.
[4] Only if **Base Derivation Key** is a DES key.
[5] Only if **Base Derivation Key** is an AES key.
[6] 32 if **Base Derivation Key** is an AES key, 16 otherwise.

Response Body:

Data	Length & Type	Description
Command Identifier	4H	**0616**.
Error Code	8H	Error code received from the HSM: **00000000** - No error code. **xxxxxxxx** - Other code defined in Error Codes.
Verification Result	1N	PIN verification result: **0** - Correct. **1** - Failed.

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3.4.6 0620 - PVV Verification ZKA

Verify a PIN from interchange using the VISA PVV method.

The [ZKA, 1.9.3.2] derivation for PIN encryption (PAC) is used.

Request Body:

Data	Length & Type	Description
Command Identifier	4H	**0620**.
PIN Key	varK	The PIN calculation key (under LMK4).
Base Derivation Key	varK	The Base derivation key (under LMK19).
RND	32H	Random Number (16-bytes).
PIN Block Format	1N	PIN block format: **0** - ISO0. **1** - ISO1. **2** - ISO2. **3** - ISO3. **4** - IBM3624. **5** - Diebold. **6** - Realsec1.
PIN Block	16	The encrypted PIN block.
PIN Extraction Method	2N	PIN extraction method: **00** - Do not use any padding value. **01** - Use digit specified in **Padding Value** field. **02** - First hexadecimal digit found in the decrypted PIN Block. **03** - Last digit found in the decrypted PIN Block. **04..16** - PIN length indicated by this value. ***Only if PIN Block Format is 4.***
PAN Length	2N	Next field length.
PAN	varN	The PAN of the card.
Padding Value	1H	The padding value use to create the PIN block. ***Only if PIN Block Format is 4 and PIN Extraction Method is 00 or 01.***
Key Index	1N	Key index digit.
PVV	4H	The PVV to verity.

Response Body:

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Data	Length & Type	Description
Command Identifier	4H	**0620**.
Error Code	8H	Error code received from the HSM: **00000000** - No error code. **xxxxxxxx** - Other code defined in Error Codes.
Verification Result	1N	PVV verification result: **0** - Correct. **1** - Failed.

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3.4.7 0621 - PIN Verification ZKA

Verify a PIN.

The [ZKA, 1.9.3.2] derivation for PIN encryption (PAC) is used.

Request Body:

Data	Length & Type	Description
Command Identifier	4H	**0621**.
PIN Key	varK	The PIN calculation key (under LMK45 (IBM German Bank Pool Institution, Interbank, ALT_1 PIN Offset or H 24), LMK46 (IBM3624 or IBM3624 PIN Offset), LMK47 (VISA PVV) or LMK4).
Base Derivation Key	varK	The base derivation key (under LMK19).
DT	16/var[1] N/H/K[2]	Decimalization table.[3]
PIN Block Format	1N	PIN block format: **0** - ISO0. **1** - ISO1. **2** - ISO2. **3** - ISO3. **4** - IBM3624. **5** - Diebold. **6** - Realsec1.
Algorithm	1N	PIN generation algorithm: **0** - IBM3624. **1** - IBM3624 PIN Offset. **4** - IBM German Bank Pool Institution. **5** - Interbank. **6** - ALT_1 PIN Offset. **7** - H24
PIN Block	16	The encrypted PIN block.
RND	32H	Random Number (16-bytes).
PIN Extraction Method	2N	PIN extraction method: **00** - Do not use any padding value. **01** - Use digit specified in **Padding Value** field. **02** - First hexadecimal digit found in the decrypted PIN Block. **03** - Last digit found in the decrypted PIN Block. **04..16** - PIN length indicated by this value. ***Only if PIN Block Format is 4.***
PAN Length	2N	Next field length.

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Data	Length & Type	Description
PAN	varN	The PAN of the card.
Padding Value	1H	The padding value use to create the PIN block. ***Only if PIN Block Format is 4 and PIN Extraction Method is 00 or 01.***
PIN Length	1N	Number of digits of the PIN. ***Only if Algorithm is 0.***
Offset Length	1N	Next field length. ***Only if Algorithm is 1, 3, 6 or 7.***
Offset	varN	The offset. ***Only if Algorithm is 1, 3, 6 or 7.***
Key Indicator	1N	Key indicator digit. ***Only if Algorithm is 5.***
Validation Field	3H	Validation data field. ***Only if Algorithm is 5.***

[1] Decimalization table length can be:
16 for a raw table.
var for a table in a container version 1.
[2] Decimalization table type can be:
N for a raw table in the clear.
H for an encrypted raw table.
K for a table in a container version 1, encrypted or in the clear.
[3] PCI PTS HSM v4.0-Approved mode of operation requires an encrypted raw table or a table in a container version 1. For new implementations, container version 1 is recommended.

Response Body:

Data	Length & Type	Description
Command Identifier	4H	**0621**.
Error Code	8H	Error code received from the HSM: **00000000** - No error code. **xxxxxxxx** - Other code defined in Error Codes.
Verification Result	1N	PIN verification result: **0** - Correct. **1** - Failed.

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3.5 PIN Translation (Cryptosec-Banking Only)

The following commands are provided to translate PIN blocks from encryption under one key to encryption under another. The commands can also translate the format of a PIN block, with the exception of those that translate to the LMK (where the PIN is not held in a standard format).

3.5.1 0603 - PIN Translation

Translate a PIN Block from encryption under a source encryption key to a destination encryption key.

Request Body:

Data	Length & Type	Description
Command Identifier	4H	**0603**.
Source PIN Transport Key	varK	Key (under LMK3) under which the PIN block is currently encrypted.
Destination PIN Transport Key	varK	Key (under LMK3) under which the PIN block is to be encrypted.
Source PIN Block Format	1N	Input PIN block format: **0** - ISO0.[1] **1** - ISO1.[1] **2** - ISO2.[1] **3** - ISO3.[1] **4** - IBM3624.[1] **5** - Diebold.[1] **6** - Realsec1.[1] **7** - ISO4.[2]
Destination PIN Block Format	1N	Output PIN block format: **0** - ISO0.[3] **1** - ISO1.[3] **2** - ISO2.[3] **3** - ISO3.[3] **4** - IBM3624.[3] **5** - Diebold.[3] **6** - Realsec1.[3] **7** - ISO4.[4]
Source PIN Block	16/32H[5]	The source PIN block encrypted under the **Source PIN Transport Key**.
PAN Length	2N	Next field length.
PAN	varN	The PAN.

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Data	Length & Type	Description
Random Length	2N	Next field length. **00..16**. ***Only if Destination PIN Block Format is 1 or 3.***
Random	varN	Random number used to form **PIN Block**. ***Only if Destination PIN Block Format is 1 or 3.***
Padding	1N	Padding value. ***Only if Destination PIN Block Format is 4.***

[1] Only if **Source PIN Transport Key** is a DES key.
[2] Only if **Source PIN Transport Key** is an AES key.
[3] Only if **Destination PIN Transport Key** is a DES key.
[4] Only if **Destination PIN Transport Key** is an AES key.
[5] 32 if **Source PIN Transport Key** is an AES key, 16 otherwise.

Response Body:

Data	Length & Type	Description
Command Identifier	4H	**0603**.
Error Code	8H	Error code received from the HSM: **00000000** - No error code. **xxxxxxxx** - Other code defined in Error Codes.
PIN Block	16/32H[1]	The translated PIN Block encrypted under the **Destination PIN Transport Key**.

[1] 32 if **Destination PIN Transport Key** is an AES key, 16 otherwise.

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3.5.2 0613 - PIN Translation with Key (3DES or AES DUKPT / ZKA)

Translate a PIN encrypted under a transpor key or a derivation key to encryption under another transport or derivation key for transmission to another node.

The following derivation methods can be used:

- ANSI X9.24-1:2009 method for TDES-DUKPT key derivation.
- ANSI X9.24-3-2017 method for AES-DUKPT key derivation.
- Zentraler Kreditausschuss (ZKA) key derivation of the German Banking Industry Committee.

Request Body:

Data	Length & Type	Description
Command Identifier	4H	**0613**.
Source Derivation Method	1N	Derivation method for the input key: **0** - None. **1** - DUKPT. **2** - ZKA.[1]
Destination Derivation Method	1N	Derivation method for the output key: **0** - None. **1** - DUKPT. **2** - ZKA.[2]
Source Encryption Key	varK	Input PIN block encryption key (under LMK3 if **Source Derivation Method** is 0, LMK19 otherwise).
Destination Encryption Key	varK	Output PIN block encryption key (under LMK3 if **Destination Derivation Method** is 0, LMK19 otherwise).
Source PIN Block Format	1N	Input PIN block format: **0** - ISO0.[1] **1** - ISO1.[1] **2** - ISO2.[1] **3** - ISO3.[1] **4** - IBM3624.[1] **5** - Diebold.[1] **6** - Realsec1.[1] **7** - ISO4.[3]

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Data	Length & Type	Description
Destination PIN Block Format	1N	Output PIN block format: **0** - ISO0.[2] **1** - ISO1.[2] **2** - ISO2.[2] **3** - ISO3.[2] **4** - IBM3624.[2] **5** - Diebold.[2] **6** - Realsec1.[2] **7** - ISO4.[4]
Source PIN Block	16/32H[5]	The input PIN block encrypted under **Source Encryption Key**.
Source PAN Length	2N	Next field length. ***Only if Source PIN Block Format is 0 or 3.***
Source PAN	varN	The PAN of **Source PIN Block**. ***Only if Source PIN Block Format is 0 or 3.***
Destination PAN Length	2N	Next field length. ***Only if Destination PIN Block Format is 0, 1 or 3.***
Destination PAN	varN	The PAN for the **Destination PIN Block**. ***Only if Destination PIN Block Format is 0, 1 or 3. If Destination PIN Block Format is 1 it shall be used to inform the random data.***
Padding Value	1H	The padding value use to create the **Destination PIN Block**. ***Only if Destination PIN Block Format is 4.***
Delimiter	1A	**;** (0x3B ASCII). Optional. If present, **Source KSN Descriptor** field shall exist. ***Only if Destination Derivation Method is 1 and Source Encryption Key is a DES key.***
Source KSN Descriptor	3N	The descriptor for the **Source KSN**. ***Only if Source Derivation Method is 1 and Source Encryption Key is a DES key.***
Source KSN	12-20H	Source Key Serial Number. ***If the Source KSN Descriptor is not present, this value shall be 20 digits in length.*** ***Only if Source Derivation Method is 1 and Source Encryption Key is a DES key.***
Source Working Key Type	2N	The type of the key to derive: **02** - AES128. **03** - AES192. **04** - AES256. ***Only if Source Derivation Method is 1 and Source Encryption Key is an AES key.***

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Data	Length & Type	Description
Source Initial Key ID	16H	The Initial Key ID. ***Only if Source Derivation Method is 1 and Source Encryption Key is an AES key.***
Source Transaction Counter	8H	The Transaction Counter. ***Only if Source Derivation Method is 1 and Source Encryption Key is an AES key.***
Delimiter	1A	**;** (0x3B ASCII). Optional. If present, **Destination KSN Descriptor** field shall exist. ***Only if Destination Derivation Method is 1 and Destination Encryption Key is a DES key.***
Destination KSN Descriptor	3N	The descriptor for the **Destination KSN**. ***Only if Destination Derivation Method is 1 and Destination Encryption Key is a DES key.***
Destination KSN	12-20H	Destination Key Serial Number. ***If the Destination KSN Descriptor is not present, this value shall be 20 digits in length.*** ***Only if Destination Derivation Method is 1 and Destination Encryption Key is a DES key.***
Destination Working Key Type	2N	The type of the key to derive: **02** - AES128. **03** - AES192. **04** - AES256. ***Only if Destination Derivation Method is 1 and Destination Encryption Key is an AES key.***
Destination Initial Key ID	16H	The Initial Key ID. ***Only if Destination Derivation Method is 1 and Destination Encryption Key is an AES key.***
Destination Transaction Counter	8H	The Transaction Counter. ***Only if Destination Derivation Method is 1 and Destination Encryption Key is an AES key.***
Source Random Data	32H	Input random data. ***Only if Source Derivation Method is 2.***
Destination Random Data	32H	Output random data. ***Only if Destination Derivation Method is 2.***

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Data	Length & Type	Description

[1] Only if **Source Encryption Key** is a DES key.
[2] Only if **Destination Encryption Key** is a DES key.
[3] Only if **Source Encryption Key** is an AES key.
[4] Only if **Destination Encryption Key** is an AES key.
[5] 32 if **Source Encryption Key** is an AES key, 16 otherwise.

Response Body:

Data	Length & Type	Description
Command Identifier	4H	**0613**.
Error Code	8H	Error code received from the HSM: **00000000** - No error code. **xxxxxxxx** - Other code defined in Error Codes.
Destination PIN Block	16/32H[1]	The output PIN block encrypted under **Destination Encryption Key**.

[1] 32 if **Destination Encryption Key** is an AES key, 16 otherwise.

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3.5.3 0617 - IBM3624 PIN Offset Translation

Translate a PIN offset from the current PIN Calculation Key to the New PIN Calculation PIN.

The new PIN Offset will be same length as the current one.

Request Body:

Data	Length & Type	Description
Command Identifier	4H	**0617**.
Current PIN Key	varK	The current PIN calculation key (under LMK46 or LMK4).
New PIN Key	varK	The new PIN calculation key (under LMK46 or LMK4).
Current DT	16/var[1] N/H/K[2]	The current decimalization table.[3]
New DT	16/var[1] N/H/K[2]	The new decimalization table.[3]
Current PAN	16H	The 16 right-most digits of the current account number. Pad with zeros on the left if needed.
New PAN	16H	The 16 right-most digits of the new account number. Pad with zeros on the left if needed.
Offset Length	2N	Next field length.
Offset	varN	The current offset.

[1] Decimalization table length can be:
16 for a raw table.
var for a table in a container version 1.
[2] Decimalization table type can be:
N for a raw table in the clear.
H for an encrypted raw table.
K for a table in a container version 1, encrypted or in the clear.
[3] PCI PTS HSM v4.0-Approved mode of operation requires an encrypted raw table or a table in a container version 1. For new implementations, container version 1 is recommended.

Response Body:

Data	Length & Type	Description
Command Identifier	4H	**0617**.
Error Code	8H	Error code received from the HSM: **00000000** - No error code. **xxxxxxxx** - Other code defined in Error Codes.
New PIN Offset	1N	The output PIN Offset.

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3.5.4 0622 - RSA Encrypted PIN Translation

Translate a PIN Block from encryption under a RSA source encryption key to a destination encryption key.

Request Body:

Data	Length & Type	Description
Command Identifier	4H	**0622**.
Source PIN Transport Key	varK	RSA Private Key (under LMK3) under which the PIN block is to be decrypted.
Destination PIN Transport Key	varK	Key (under LMK3) under which the PIN block is to be encrypted.
Algorithm	2N	RSA decryption algorithm: **01** - RSA-PKCS#1 v1.5. **02** - RSA-OAEP.
MGF	2N	Mask Generation Function: **01** - MGF1. ***Only if Algorithm is 02.***
Hash Algorithm	1A	Hash algorithm used to calculate the digest of the RSA-OAEP encoding parameters: **0** - SHA-1 **1** - SHA-224 **2** - SHA-256 **3** - SHA-384 **4** - SHA-512 **5** - SHA-512/224 **6** - SHA-512/256 ***Only if Algorithm is 02.***
MGF Hash Algorithm	1A	RSA-OAEP MGF hash function: **0** - SHA-1 **1** - SHA-224 **2** - SHA-256 **3** - SHA-384 **4** - SHA-512 **5** - SHA-512/224 **6** - SHA-512/256 ***Only if Algorithm is 02.***
Encoding Parameters Length	6N	Next field length. ***Only if Algorithm is 02.***
Encoding Parameters	varH	Encoding Parameters. ***Only if Algorithm is 02 and Encoding Parameters Length is not 000000.***

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Data	Length & Type	Description
Source PIN Block Length	6N	Next field length.
Source PIN Block	varH	The source PIN block encrypted under the **Source PIN Transport Key**.
Source PIN Block Format	1N	Input PIN block format: **0** - ISO0. **1** - ISO1. **2** - ISO2. **3** - ISO3. **4** - IBM3624. **5** - Diebold.
Destination PIN Block Format	1N	Output PIN block format: **0** - ISO0. **1** - ISO1. **2** - ISO2. **3** - ISO3. **4** - IBM3624. **5** - Diebold.
PAN Length	2N	Next field length.
PAN	varN	The PAN.

Response Body:

Data	Length & Type	Description
Command Identifier	4H	**0622**.
Error Code	8H	Error code received from the HSM: **00000000** - No error code. **xxxxxxxx** - Other code defined in Error Codes.
PIN Block	16H	The translated PIN Block encrypted under the **Destination PIN Transport Key**.

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3.5.5 1803 - LMK Encrypted PIN Block Translate

Translate a PIN block protected by the LMK3, from the Old Master Keys to the current Master Keys.

Request Body:

Data	Length & Type	Description
Command Identifier	4H	**1803**.
PIN Block Format	1N	PIN block format: **0** - ISO0. **1** - ISO1. **2** - ISO2. **3** - ISO3. **4** - IBM3624. **5** - Diebold. **6** - Realsec1. **7** - ISO4.
PIN Block	16H/32H[1]	The input PIN block. The Realsec1 PIN block shall be padded with '0' to the right.
PAN Length	2N	Next field length. ***Only if PIN Block Format is 0, 3 or 6.***
PAN	varN	The PAN of **PIN Block**. ***Only if PIN Block Format is 0, 3 or 6.***
Padding Value	1H	The padding value use to create the **Destination PIN Block**. ***Only if PIN Block Format is 4.***
RND Length	2N	Next field length. ***Only if PIN Block Format is 1 or 3.***
RND	varN	Random data. No data for internal random nuber generation. ***Only if PIN Block Format is 1 or 3.***

[1] 32 if **PIN Block Format** is ISO4. 16 otherwise.

Response Body:

Data	Length & Type	Description
Command Identifier	4H	**1803**.
Error Code	8H	Error code received from the HSM: **00000000** - No error code. **xxxxxxxx** - Other code defined in Error Codes.
Destination PIN Block	var N/H[1]	The output PIN block.

[1] 32H if **PIN Block Format** is ISO4. Realsec1 PIN Block Length + 1 N if **PIN Block Format** is Realsec1. 16H otherwise.

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3.6 Validation Codes Calculation and Verification (Cryptosec-Banking Only)

3.6.1 0701 - CVV/CVC/CSS Calculation and Verification

This command can also be used to verify or generate:

- The VISA CVV1 or MasterCard CVC1 for encoding onto the card's magnetic stripe.
- VISA CVV2 or Mastercard CVC2 for printing on a card's signature strip. In this case, the **Service Code** field shall be set to **000**.
- VISA iCVV for inclusion on a Visa EMV chip card. In this case, the **Service Code** field shall be set to **999**.
- VISA CAAV or Mastercard AVV for use in the 3-D Secure environment.

Request Body:

Data	Length & Type	Description
Command Identifier	4H	**0701**.
CVV/CVC Key	varK	CVV/CVC/CSS key (under LMK5, LMK29, LMK35 or LMK6).
Expiration Date	4N	Expiration date.
Service Code	3N	Service code.
Delimiter	1A	**;** (0x3B ASCII). Optional. If present, **PAN Length** field shall exist.
PAN Length	2N	Next field length. **12..19**.
PAN	16/varN	The personal account number. ***If PAN Length is not present, this value shall be 16 digits in length.***
Operation	1N	Mode: **0** - Generate. **1** - Verify.
Validation Value/Code	3N	Validation value/code to verify. ***Only if Mode is 1.***

Response Body:

Data	Length & Type	Description
Command Identifier	4H	**0701**.

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Data	Length & Type	Description
Error Code	8H	Error code received from the HSM: **00000000** - No error code. **xxxxxxxx** - Other code defined in Error Codes.
Validation Value/Code	3N	Computed validation code. ***Only if Mode is 0.***
Validation Result	1N	Validation result: **0** - Correct. **1** - Failed. ***Only if Mode is 1.***

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3.6.2 0703 - CSC Calculation and Verification

Generate an American Express Card Security Code (CSC).

This command calculates the 3, 4 and 5-digit CSC Values from the supplied data according to:

- Classic CSC Algorithm (CSC Version 1.0).
- Enhanced CSC Algorithm (CSC Version 2.0).

Request Body:

Data	Length & Type	Description
Command Identifier	4H	**0703**.
CSC Key	varK	CSC key (under LMK 5).
Mode	1N	Operation mode: **0** - Calculation. **1** - Validation.
PAN	16N	16 left most digits of the Personal Account Number.[1] If the PAN is shorter than 16 characters, it shall be right padded with zeros '0'.
Expiration Date	4N	The expiration date of the card.
Delimiter	1A	**;** (0x3B ASCII). Optional. If present, **Service Code** field shall exist. ***Mandatory for Enhanced CSC Algorithm (CSC Version 2.0).***
Service Code	3N	The Service Code. ***Mandatory for Enhanced CSC Algorithm (CSC Version 2.0).***
CSC Length	2N	Next field length.[2] ***Only if Mode is 1.***
CSC3||CSC4||CSC5	varH	CSC to verify. If any of the values don't need to be verified, type them as 'F's. Example: If CSC4 and CSC5 need to be verified, pass FFF||CSC4||CSC5. If just CSC4 needs to be verified, pass FFF||CSC4||FFFFF or just the CSC4 value. ***Only if Mode is 1.***

[1] It should start with '34' or '37'.
[2] Only valid values: '03', '04', '05' and '12'.

Response Body:

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Data	Length & Type	Description
Command Identifier	4H	**0703**.
Error Code	8H	Error code received from the HSM: **00000000** - No error code. **xxxxxxxx** - Other code defined in Error Codes.
VER	1N	Validation result: **0** - Correct. **1** - Failed. ***Only if Mode is 1.***
CSC3	3N	Computed CSC3. ***Only if Mode is 0.***
CSC4	4N	Computed CSC4. ***Only if Mode is 0.***
CSC5	5N	Computed CSC5. ***Only if Mode is 0.***

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3.6.3 0704 - Verify a Dynamic CVV/CVC

Verify or generate a Dynamic Card Verification Value (dCVV) or Card Verification Code (dCVC) as well as Mastercard's CVC3.

The PAN sequence number is assumed to be zero for the calculation of the UDK. This command will append a zero to the supplied PAN when generating the DK-AC.

The EMV "Track 2 Equivalent Data", provided in the authorization message and originating from the contactless smart card, is the source for the following data fields:

- PAN.
- Expiration Date.
- Service Code.
- ATC.
- DCVV.

The PAN sequence number is assumed to be zero for the calculation of the DK-CVC3. This command will append a zero to the supplied PAN when generating the DK-CVC3. Optionally, the command will also allow the PAN sequence number to be provided so that the calculation of the UDK can be based on supplied PAN and PAN sequence numbers.

Verification of the CVC3 requires the IVCVC3 which is a MAC calculated over the static part of Track 1 or Track 2 data using the DK-CVC3. This command offers the option of providing this parameter for verification of the CVC3 or it can create the IVCVC3 during the verification process if the appropriate Track Data is provided.

PINCVC3 is a variant of the standard CVC3 scheme. A Mobile MasterCard PayPass M/Chip application generates a PINCVC3 instead of a CVC3 when the PIN has been entered on the mobile device. A different IV (known as a PINIVCVC3) is used in this calculation. The calculation of the PINCVC3 is otherwise identical to the CVC3.

To assist in development of host applications, the calculated Dynamic CVV will be provided in the response data if Error Code 01 is returned and the HSM is in Authorized State.

Request Body:

Data	Length & Type	Description
Command Identifier	4H	**0704**.
CVC Key	varK	CVV/CVC/CSS key (under LMK5, LMK29, LMK35 or LMK6).

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Data	Length & Type	Description
Version	1N	Version of the CVC3 to verify: **0** - Visa dCVV. **1** - MasterCard CVC3. IVCVC3 specify in input and PSN=00. **2** - MasterCard CVC3. IVCVC3 and PSN specify in input. **3** - MasterCard CVC3. IVCVC3 calculated from provided Track Data and PSN specified in input.
Derivation Method	1N	Method to get ICC MK from the **CVV Key**: **1** - EMV 4.x Option A ICC Master Key Derivation.
PAN Length	2N	
PAN	varN	The PAN of the card.
PSN	2N	The PAN Sequence number of the card. ***Only if Version is 2 or 3.***
Expiration Date	4N	The Card's expiration date. ***Only if Version is 0.***
Service Code	3N	The card's track2 service code. ***Only if Version is 0.***
ATC	6N	The Application Transaction Counter provided on track2. If it is less than 6 digits, it should be right justified and padded on the left with zeros. ***Only if Version is 0.***
DCVV	3N	Dynamic CVV to be verified. ***Only if Version is 0.***
IVCVC3	4H	The issuer propietary static data. ***Only if Version is 1 or 2.***
Track Data Length	3N	Next field length. ***Only if Version is 3.***
Track Data	varH	Static Track (1 or 2) Data. ***Only if Version is 3.***
UN	8H	Random number provided to the card by the terminal during the transaction. ***Only if Version is 1, 2 or 3.***
Card ATC	4H	The ATC from the card. ***Only if Version is 1, 2 or 3.***
CVC3 Length	1N	Next field length. **3** or **5**. ***Only if Version is 1, 2 or 3.***
CVC3	varN	Last n digits of the dynamic CVC3 to verify. ***Only if Version is 1, 2 or 3.***

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Response Body:

Data	Length & Type	Description
Command Identifier	4H	**0704**.
Error Code	8H	Error code received from the HSM: **00000000** - No error code. **xxxxxxxx** - Other code defined in Error Codes.
Validation Result	1N	Validation result: **0** Correct. **1** Failed.
Diagnostic Data	3N or 5N	Computed CVC3. ***Only if Validation failed and HSM is in Authorization mode.***

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3.7 Message Securization

The Cryptosec-Banking HSM provides commands to generate, verify and translate a MAC. Because the HSM has no flow control, the application programmer is responsible for ensuring that the input buffer is not exceeded. The HSM input buffer is approximately 999 KB per connection in length. The length of the input buffer limits the amount of data over which a MAC can be calculated in a single call to the HSM.

3.7.1 0801 - Generate or Verify MAC

Generate or verify the MAC of a message.

Request Body:

Data	Length & Type	Description
Command Identifier	4H	**0801**.
MAC Key	varK	MAC key (under LMK37 (CBC MAC, MAC TDES, MAC ANSI X9.9-1), LMK39 (MAC ANSI X9.19-1), or LMK8).
Algorithm	1N	Algorithm indicator: **0** - MAC ANSI X9.9-1. **1** - MAC ANSI X9.19-1. **2** - MAC TDES. **3** - CBC MAC (AES only).
Padding	1N	Padding method: **0** - ISO/IEC 9797 method 1 (pad with 0x00 if necessary). **1** - ISO/IEC 9797 method 2 (add 0x80 and pad with 0x00 if necessary). **2** - ISO/IEC 9797 method 3 (prepend message with its length then pad with 0x00 if necessary).
Hash	1N	Previous hash: **0** - No Hash. **1** - SHA-1. **2** - MD5. **3** - SHA-224. **4** - SHA-256. **5** - SHA-384. **6** - SHA-512. **7** - SHA-512/224. **8** - SHA-512/256.
Initialization Vector	16/32H[1]	Initialization vector.
Message Length	6N	Next field length.

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Data	Length & Type	Description
Message	varH	Data over which the MAC is to be performed.
MAC to Verify	16/32H[1]	MAC to verify. ***Only for MAC verification.***

[1] 16 for DES keys and 32 for AES keys.

Response Body:

Data	Length & Type	Description
Command Identifier	4H	**0801**.
Error Code	8H	Error code received from the HSM: **00000000** - No error code. **xxxxxxxx** - Other code defined in Error Codes.
Verification Result	1N	MAC verification result: **0** - Correct. **1** - Failed. ***Only if MAC to Verify was provided.***
Calculated MAC	16/32H[1]	Calculated MAC. ***Only if MAC to Verify was not provided.***

[1] 16 for DES keys and 32 for AES keys.

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3.7.2 0802 - Generate or Verify CMAC

Generate or authenticate a CMAC of a message.

Request Body:

Data	Length & Type	Description
Command Identifier	4H	**0802**.
MAC	varK	AES MAC key (under LMK42 or LMK8).
Delimiter	1A	**;** (0x3B ASCII). Optional. If present, **Padding** field shall exist. Otherwise ISO 9797 method 2 will be applied to the message.
Padding	1N	Padding method: **0** - ISO/IEC 9797 method 1 (pad with 0x00 if necessary). **1** - ISO/IEC 9797 method 2 (add 0x80 and pad with 0x00 if necessary). **2** - ISO/IEC 9797 method 3 (prepend message with its length then pad with 0x00 if necessary). **3** - No padding.
Delimiter	1A	**#** (0x23 ASCII). Optional. If present, **Hash** field shall exist. Otherwise No Hash will be performed to the message.
Hash	1N	Previous hash: **0** - No Hash. **1** - SHA-1. **2** - MD5. **3** - SHA-224. **4** - SHA-256. **5** - SHA-384. **6** - SHA-512. **7** - SHA-512/224. **8** - SHA-512/256.
Initialization Vector	32H	Initialization vector.
Message Length	6N	Next field length.
Message	varH	Data over which the MAC is to be performed.
CMAC to Verify	32H	CMAC to verify. ***Only for CMAC verification.***

Response Body:

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Data	Length & Type	Description
Command Identifier	4H	**0802**.
Error Code	8H	Error code received from the HSM: **00000000** - No error code. **xxxxxxxx** - Other code defined in Error Codes.
Verification Result	1N	CMAC verification result: **0** - Correct. **1** - Failed. ***Only if CMAC to Verify was provided.***
CMAC	32H	Generated CMAC. ***Only if CMAC to Verify was not provided.***

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3.7.3 0803 - Generate or Verify HMAC

Generate or verify an HMAC of a message.

Request Body:

Data	Length & Type	Description
Command Identifier	4H	**0803**.
MAC Key	varK	MAC key (under LMK43 or LMK8).
Hash	1N	Hash function: **0** - MD5. **1** - SHA-1. **2** - SHA-224. **3** - SHA-256. **4** - SHA-384. **5** - SHA-512. **6** - SHA-512/224. **7** - SHA-512/256.
Message Length	6N	Next field length.
Message	varH	Data over which the HMAC is to be performed.
HMAC to Verify Length	6N	Length in bytes of the HMAC to be verified (next field). Shall be between half of the size of the Hash Algorithm selected and its full size. For example, if it's chosen SHA-1 (20 bytes = 40 hexa chars), this field shall be between 20 and 40 and shall be multiple of 2. ***Only for HMAC verification.***
HMAC to Verify	varH	The HMAC to be verified. ***Only for HMAC verification.***

Response Body:

Data	Length & Type	Description
Command Identifier	4H	**0803**.
Error Code	8H	Error code received from the HSM: **00000000** - No error code. **xxxxxxxx** - Other code defined in Error Codes.

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Data	Length & Type	Description
Verification Result	1N	HMAC verification result: **0** - Correct. **1** - Failed. ***Only if HMAC to Verify was provided.***
HMAC	varH[1]	Generated HMAC. ***Only if HMAC to Verify was not provided.***

[1] Its size is the digest size of the hash function selected.

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3.7.4 0804 - Generate or Verify MAC using a Derived Key (Cryptosec-Banking Only)

Generate or verify the MAC of a message.

The following derivation methods can be used:

- ANSI X9.24-1:2009 method for DUKPT key derivation.
- Zentraler Kreditausschuss (ZKA) key derivation of the German Banking Industry Committee.

Request Body:

Data	Length & Type	Description
Command Identifier	4H	**0804**.
Derivation Method	1N	Derivation method: **2** - 3DES DUKPT. **3** - ZKA.
Base Derivation Key	varK	The base derivation key (under LMK19).
Delimiter	1A	**;** (0x3B ASCII). Optional. If present, **KSN Descriptor** field shall exist. ***Only if Derivation Method is 2.***
KSN Descriptor	3N	The descriptor for the **KSN**. ***Only if Derivation Method is 2.***
KSN	12-20H	Current Key Serial Number. ***If the KSN Descriptor is not present, this value shall be 20 digits in length.*** ***Only if Derivation Method is 2.***
Random Data	32H	Random data. ***Only if Derivation Method is 3.***
Algorithm	1N	Algorithm indicator: **1** - MAC ANSI X9.19-1. **2** - MAC TDES.
Padding	1N	Padding method: **0** - ISO/IEC 9797 method 1 (pad with 0x00 if necessary). **1** - ISO/IEC 9797 method 2 (add 0x80 and pad with 0x00 if necessary). **2** - ISO/IEC 9797 method 3 (prepend message with its length then pad with 0x00 if necessary).

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Data	Length & Type	Description
Hash	1N	Previous hash: **0** - No Hash. **1** - SHA-1. **2** - MD5. **3** - SHA-224. **4** - SHA-256. **5** - SHA-384. **6** - SHA-512. **7** - SHA-512/224. **8** - SHA-512/256.
Initialization Vector	16H	Initialization vector. ***Shall always be present but it is only used when Algorithm is 2.***
Message Length	6N	Next field length.
Message	varH	Data over which the MAC is to be performed.
MAC to Verify Length	6N	Next field length. Values: **000008**, **000012** or **000016**. ***Only for MAC verification.***
MAC to Verify	varH	MAC to verify. ***Only for MAC verification.***

Response Body:

Data	Length & Type	Description
Command Identifier	4H	**0804**.
Error Code	8H	Error code received from the HSM: **00000000** - No error code. **xxxxxxxx** - Other code defined in Error Codes.
Verification Result	1N	MAC verification result: **0** - Correct. **1** - Failed. ***Only if MAC to Verify was provided.***
Calculated MAC	16H	Calculated MAC. ***Only if MAC to Verify was not provided.***

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3.7.5 0805 - HMAC Initialization

Initialize an HMAC.

Request Body:

Data	Length & Type	Description
Command Identifier	4H	**0805**.
MAC Key	varK	MAC key (under LMK43 or LMK8).
Hash	1N	Hash function: **0** - MD5. **1** - SHA-1. **2** - SHA-224. **3** - SHA-256. **4** - SHA-384. **5** - SHA-512. **6** - SHA-512/224. **7** - SHA-512/256.

Response Body:

Data	Length & Type	Description
Command Identifier	4H	**0805**.
Error Code	8H	Error code received from the HSM: **00000000** - No error code. **xxxxxxxx** - Other code defined in Error Codes.
Output State	varK	State.[1]

[1] Ephemeral object to be used as input to a subsequent 0806 or 0807 command. Its size and format can be changed without notice.

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3.7.6 0806 - HMAC Update

Update an HMAC.

Request Body:

Data	Length & Type	Description
Command Identifier	4H	**0806**.
Input State	varK	State.[1]
Message Length	6N	Next field length.
Message	varH	Data over which the HMAC is to be performed.

[1] Ephemeral object output from a previous 0805 or 0806 command. Its size and format can be changed without notice.

Response Body:

Data	Length & Type	Description
Command Identifier	4H	**0806**.
Error Code	8H	Error code received from the HSM: **00000000** - No error code. **xxxxxxxx** - Other code defined in Error Codes.
Output State	varK	State.[1]

[1] Ephemeral object to be used as input to a subsequent 0806 or 0807 command. Its size and format can be changed without notice.

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3.7.7 0807 - HMAC Finalize

Finalize an HMAC generation or verification.

Request Body:

Data	Length & Type	Description
Command Identifier	4H	**0807**.
Input State	varK	State.[1]
Message Length	6N	Next field length.
Message	varH	Data over which the HMAC is to be performed.
HMAC to Verify Length	6N	Length in bytes of the HMAC to be verified (next field). Shall be between half of the size of the Hash Algorithm selected and its full size. For example, if it's chosen SHA-1 (20 bytes = 40 hexa chars), this field shall be between 20 and 40 and shall be multiple of 2. ***Only for HMAC verification.***
HMAC to Verify	varH	The HMAC to be verified. ***Only for HMAC verification.***

[1] Ephemeral object output from a previous 0805 or 0806 command. Its size and format can be changed without notice.

Response Body:

Data	Length & Type	Description
Command Identifier	4H	**0807**.
Error Code	8H	Error code received from the HSM: **00000000** - No error code. **xxxxxxxx** - Other code defined in Error Codes.
Verification Result	1N	HMAC verification result: **0** - Correct. **1** - Failed. ***Only if HMAC to Verify was provided.***
HMAC	varH[1]	Generated HMAC. ***Only if HMAC to Verify was not provided.***

[1] Its size is the digest size of the hash function selected on the associated 0805 command.

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3.8 Data Encryption

3.8.1 0903 - AES or 3DES Encryption and Decryption

AES or 3DES encryption and decryption.

It is the responsability of the host to send the precise amount of data. For encryption, no extra padding will be added to the message.

Request Body:

Data	Length & Type	Description
Command Identifier	4H	**0903**.
Key	varK	Encryption/Decryption key (under LMK7).
Operation	1N	Operation to perform: **0** - Hexadecimal decryption. **1** - Hexadecimal encryption. **2** - Binary decryption. **3** - Binary encryption.
Mode	1N	Mode: **0** - CBC. **1** - ECB.
Initialization Vector	16/32H[1]	Initialization vector. ***Only if Mode is 0.***
Message Length	6N	Next field length.[2]
Message	varH	The data to encrypt or decrypt.

[1] 16 for DES keys and 32 for AES keys.
[2] If **Operation** is 0 or 1 **Message Length** shall be multiple of 16 for 3DES keys or 32 for AES keys.
If **Operation** is 2 or 3 **Message Length** shall be multiple of 8 for 3DES keys and 16 for AES keys.

Response Body:

Data	Length & Type	Description
Command Identifier	4H	**0903**.
Error Code	8H	Error code received from the HSM: **00000000** - No error code. **xxxxxxxx** - Other code defined in Error Codes.

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Data	Length & Type	Description
Output Data Length	6N	Next field length.
Output Data	varH	Encrypted or decrypted data.

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3.8.2 0910 - AES Encryption and Decryption - PBKDF2

AES encryption and decryption with PBKDF2 key generation algorithm.

It is the responsability of the host to send the precise amount of data. For encryption, no extra padding will be added to the message.

Request Body:

Data	Length & Type	Description
Command Identifier	4H	**0910**.
Direction	1N	Operation to perform: **0** - Decryption. **1** - Encryption.
Format	1N	Format of the data to be informed: **0** - Binary or Text. **1** - Hexadecimal.
Mode	1N	Mode: **0** - CBC. **1** - ECB.
Initialization Vector	16/32H[1]	Initialization vector. ***Only if Mode is 0.***
Key Length	4N	Length in bits of the key to generate. It will take the following values: - **0128**, **0192** or **0256**.
Password Length	4N	Next field length.[2]
Password	varB	The master password from which the derived key is generated.
Salt Length	2N	Next field length.[3]
Salt	varB	The salt.
Interation Count	6N	number of iterations desired.[4]
Message Length	6N	Next field length.[5]
Message	varH	The data to encrypt or decrypt.

[1] 16 for DES keys and 32 for AES keys.
[2] It should be up to 64 bytes long.
[3] It should be at least 16 bytes long.
[4] A minimum of 1000 is recommended.
[5] If **Format** is 0 **Message Length** shall be multiple of 16 bytes.
If **Format** is 1 **Message Length** shall be multiple of 32 characters.

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Response Body:

Data	Length & Type	Description
Command Identifier	4H	**0910**.
Error Code	8H	Error code received from the HSM: **00000000** - No error code. **xxxxxxxx** - Other code defined in Error Codes.
Output Data Length	6N	Next field length.
Output Data	varH	Encrypted or decrypted data.

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3.8.3 0911 - Symmetric Encryption and Decryption TDES DUKPT

Symmetric encryption and decryption with DUKPT derivation.

It is responsible of the host to send the precise amount of data. For encryption, no extra padding will be added to the message.

Request Body:

Data	Length & Type	Description
Command Identifier	4H	**0911**.
Direction	1N	Operation to perform: **0** - Encryption. **1** - Decryption.
Mode	1N	Encryption/Decryption mode to use: **0** - ECB. **1** - CBC.
Initialization Vector	16H	Initialization vector. ***Only if Mode is 1.***
Base Derivation Key	varK	Base derivation key (under LMK19).
KSN Descriptor	3H	The descriptor for the KSN (next field).
KSN	12-20H	The Key Serial Number sent by the PIN pad.
Message Length	6N	Next field length. ***Shall be multiple of 16.***
Message	varH	The message to encrypt or decrypt.

Response Body:

Data	Length & Type	Description
Command Identifier	4H	**0911**.
Error Code	8H	Error code received from the HSM: **00000000** - No error code. **xxxxxxxx** - Other code defined in Error Codes.
Output Data Length	6N	Next field length.
Output Data	varH	The message encrypted or decrypted.

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3.8.4 0912 - AES-GCM Encryption and Decryption

AES GCM encryption and decryption.

It is responsibility of the host to send the precise amount of data. For encryption, no extra padding will be added to the message.

Request Body:

Data	Length & Type	Description
Command Identifier	4H	**0912**.
Direction	1N	Operation to perform: **0** - Decryption. **1** - Encryption.
Format	1N	Format of the data to be informed: **0** - Binary or Text. **1** - Hexadecimal.
Operational Key	varK	Encryption/Decryption key (under LMK7).
Initialization Vector Length	6N	Next field length.
Initialization Vector	varB/H	Initialization vector.[1] ***Shall always exists, but it will only be used if the HSM is not configured in NIST FIPS PUB 140-2-Approved mode.***
Additional Authentication Data Length	6N	Next field length.[2]
Additional Authentication Data	varB/H	The Additional Authentication Data.[2]
Authentication Tag Length	2N	Next field length.[3] Valid Lengths: If **Format** is **0**: **04**, **08**, **12**, **13**, **14**, **15** or **16**. If **Format** is **1**: **08**, **16**, **24**, **26**, **28**, **30** or **32**.
Authentication Tag	varB/H	Authentication Tag. ***Only if Direction is 0.***
Message Length	6N	Next field length.[2]
Message	varB/H	The data to encrypt or decrypt.[2]

[1] **12** or greater if **Format** is **0** or **24** or greater if **Format** is **1**.
[2] If **Format** is **0**, any value shall be valid. It shall be used as received.
If **Format** is **1**, it shall be multiple of 2. It shall be compressed before being used.
[3] If **Direction** is **0** Length of the Authentication Tag to be authenticated previous data decryption.
If **Direction** is **1** Length of the Authentication Tag to be returned along with the encrypted data.

Response Body:

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Data	Length & Type	Description
Command Identifier	4H	**0912**.
Error Code	8H	Error code received from the HSM: **00000000** - No error code. **xxxxxxxx** - Other code defined in Error Codes.
Authentication Tag Length	2N	Next field length. ***Only if Direction is 1.***
Authentication Tag	varB/H	Authentication Tag.[1] ***Only if Direction is 1.***
Output Data Length	6N	Next field length.
Output Data	varB/H	Output data.

[1] Same length as specified at **Authentication Tag Length**.

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3.8.5 1003 - RSA Encryption and Decryption

RSA encryption and decryption.

Request Body:

Data	Length & Type	Description
Command Identifier	4H	**1003**.
RSA Key	varK	The key to use (under LMK 60).
Direction	1N	Direction: **0** - Encrypt with public key. **1** - Decrypt with private key.
Algorithm	1N	Algorithm: **0** - RSA-PKCS#1 v1.5. **1** - RSA-OAEP.
Hash Algorithm	1A	Hash function: **0** - SHA-1 **1** - SHA-224 **2** - SHA-256 **3** - SHA-384 **4** - SHA-512 **5** - SHA-512/224 **6** - SHA-512/256 ***Only if Algorithm is 1.***
MGF1 Hash Algorithm	1A	MGF1 hash function: **0** - SHA-1 **1** - SHA-224 **2** - SHA-256 **3** - SHA-384 **4** - SHA-512 **5** - SHA-512/224 **6** - SHA-512/256 ***Only if Algorithm is 1.***
Message Length	6N	Next field length.
Message	varH	The message to encrypt or decrypt.

Response Body:

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Data	Length & Type	Description
Command Identifier	4H	**1003**.
Error Code	8H	Error code received from the HSM: **00000000** - No error code. **xxxxxxxx** - Other code defined in Error Codes.
Output data Length	6N	Next field length.
Output data	varH	The output data.

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3.9 Data Signature

3.9.1 1103 - Asymmetric Signature

Generate an asymmetric signature of a message.

Take into account the following guidelines regarding RSA-PKCS#1 v2.2 (RSASSA-PSS) signatures:

- No hash identifier is not allowed.
- The hash function selected is also used on the MGF.
- The MGF used is always MGF1.
- The salt length is automatically defined as the length of the output of the selected hash function. In order to define an arbitrary salt length, please use the 1105 command.

Request Body:

Data	Length & Type	Description
Command Identifier	4H	**1103**.
RSA Key	varK	Signature key (under LMK61 or LMK62). ***LMK62 only allowed when the HSM is not configured in PCI PTS HSM v4.0-Approved mode.***
Hash Identifier	2N	Hash function: **00** - No hash. **01** - MD5. **10** - SHA-1. **11** - SHA-224. **12** - SHA-256. **13** - SHA-384. **14** - SHA-512.
Mechanism	2N	Signature mechanism: **01** - RSA.
Algorithm	2N	Signature algorithm: **00** - Raw. **01** - RSA-PKCS#1 v1.5. **02** - RSA-PKCS#1 v2.2 (RSASSA-PSS). **10** - X9.31.
Hashed Message Length	6N	Next field length.

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Data	Length & Type	Description
Hashed Message	varH	Hashed message. ***Data already hashed as indicated by Hash Identifier field. The length of the message shall be the length of the output of the selected hash function. When no hash is selected, the length of the message is up to the modulus length of the key.***

Response Body:

Data	Length & Type	Description
Command Identifier	4H	**1103**.
Error Code	8H	Error code received from the HSM: **00000000** - No error code. **xxxxxxxx** - Other code defined in Error Codes.
Signature Length	6N	Next field length.
Signature	varH	Signature.

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3.9.2 1104 - Asymmetric Signature Verification

Verify an asymmetric signature.

Take into account the following guidelines regarding RSA-PKCS#1 v2.2 (RSASSA-PSS) signatures:

- No hash identifier is not allowed.
- The hash function selected is also used on the MGF.
- The MGF used is always MGF1.
- The salt length is automatically defined as the length of the output of the selected hash function. In order to define an arbitrary salt length, please use the 1106 command.

Request Body:

Data	Length & Type	Description
Command Identifier	4H	**1104**.
RSA Key	varK	Signature key (under LMK61 or LMK62). ***LMK62 only allowed when the HSM is not configured in PCI PTS HSM v4.0-Approved mode.***
Hash Identifier	2N	Hash function: **00** - No hash. **01** - MD5. **10** - SHA-1. **11** - SHA-224. **12** - SHA-256. **13** - SHA-384. **14** - SHA-512.
Mechanism	2N	Signature mechanism: **01** - RSA.
Algorithm	2N	Signature algorithm: **00** - Raw. **01** - RSA-PKCS#1 v1.5. **02** - RSA-PKCS#1 v2.2 (RSASSA-PSS). **10** - X9.31.
Hashed Message Length	6N	Next field length.

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Data	Length & Type	Description
Hashed Message	varH	Hashed message. ***Data already hashed as indicated by Hash Identifier field. The length of the message shall be the length of the output of the selected hash function. When no hash is selected, the length of the message is up to the modulus length of the key.***
Signature Length	6N	Next field length.
Signature	varH	Signature.

Response Body:

Data	Length & Type	Description
Command Identifier	4H	**1104**.
Error Code	8H	Error code received from the HSM: **00000000** - No error code. **xxxxxxxx** - Other code defined in Error Codes.
Verification Result	1N	Verification result: **0** - Correct. **1** - Failed.

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3.9.3 1105 - RSA-PKCS#1 v2.2 (RSASSA-PSS) Signature

Generate a RSA-PKCS#1 v2.2 (RSASSA-PSS) signature of a message.

Note: The hash function selected is also used on the MGF.

Request Body:

Data	Length & Type	Description
Command Identifier	4H	**1105**.
RSA Key	varK	Signature key (under LMK61 or LMK62). ***LMK62 only allowed when the HSM is not configured in PCI PTS HSM v4.0-Approved mode.***
Hash Identifier	2N	Hash function: **01** - MD5. **10** - SHA-1. **11** - SHA-224. **12** - SHA-256. **13** - SHA-384. **14** - SHA-512.
MGF Identifier	2N	Mask Generation function: **01** - MGF1.
Salt Length	6N	Salt length.
Hashed Message	varH	Hashed message. ***Data already hashed as indicated by Hash Identifier field. The length of the message shall be the length of the output of the selected hash function.***

Response Body:

Data	Length & Type	Description
Command Identifier	4H	**1105**.
Error Code	8H	Error code received from the HSM: **00000000** - No error code. **xxxxxxxx** - Other code defined in Error Codes.
Signature Length	6N	Next field length.
Signature	varH	Signature.

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3.9.4 1106 - RSA-PKCS#1 v2.2 (RSASSA-PSS) Signature Verification

Verify a RSA-PKCS#1 v2.2 (RSASSA-PSS) signature of a message.

Note: The hash function selected is also used on the MGF.

Request Body:

Data	Length & Type	Description
Command Identifier	4H	**1106**.
RSA Key	varK	Signature key (under LMK61 or LMK62). ***LMK62 only allowed when the HSM is not configured in PCI PTS HSM v4.0-Approved mode.***
Hash Identifier	2N	Hash function: **01** - MD5. **10** - SHA-1. **11** - SHA-224. **12** - SHA-256. **13** - SHA-384. **14** - SHA-512.
MGF Identifier	2N	Mask Generation function: **01** - MGF1.
Salt Length	6N	Salt length.
Hashed Message	varH	Hashed message. ***Data already hashed as indicated by Hash Identifier field. The length of the message shall be the length of the output of the selected hash function.***
Signature Length	6N	Next field length.
Signature	varH	Signature.

Response Body:

Data	Length & Type	Description
Command Identifier	4H	**1106**.
Error Code	8H	Error code received from the HSM: **00000000** - No error code. **xxxxxxxx** - Other code defined in Error Codes.

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Data	Length & Type	Description
Verification Result	1N	Verification result: **0** - Correct. **1** - Failed.

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3.10 Printing

The Cryptosec-Banking HSM can print PIN mailers (the contents of which can be read only after the form has been opened) and other types of data that shall be kept secret.

PIN mailers can be used to send PINs to cardholders. Also, if the cardholder is to be given the opportunity of selecting his/her own PIN by mail (instead of at an entry device), solicitation data can be sent; it is not necessary to send a PIN if only a solicitation request is to be sent.

To configure the printer, use the HSM's console.

3.10.1 0609 - PIN Printing (Cryptosec-Banking Only)

Print the PIN/PIN and solicitation data at the HSM-attached terminal.

Request Body:

Data	Length & Type	Description
Command Identifier	4H	**0609**.
PIN Encryption Key	varK	PIN Block transport key (under LMK3).
PIN Block Format	1N	Input PIN block format: **0** - ISO0.[1] **1** - ISO1.[1] **2** - ISO2.[1] **3** - ISO3.[1] **4** - IBM3624.[1] **5** - Diebold.[1] **6** - Realsec1.[1] **7** - ISO4.[2]
PIN Block	16/32H[3]	Encrypted PIN block.
PAN Length	2N	Next field length.
PAN	varN	The account number.
User Data Length	4N	Next field length.
User Data	varA	Variable user data table (name, custodian, ATM, dates, uses, key…). Format will depend on printing format already defined.

[1] Only if **PIN Block transport key** is a DES key.
[2] Only if **PIN Block transport key** is an AES key.
[3] 32 if **PIN Block transport key** is an AES key, 16 otherwise.

Response Body:

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Data	Length & Type	Description
Command Identifier	4H	**0609**.
Error Code	8H	Error code received from the HSM: **00000000** - No error code. **xxxxxxxx** - Other code defined in Error Codes.

3.10.1.1 PIN Printing

It is necessary to define the printing template by running the command 1301 before performing printing operation.

In PIN printing, the argument **T** contains print strings 0 to 15 that are referenced in the template. There is no need to define all of the 16 strings. However, it shall be taken into account that they are numbered in a sequential manner starting from 0. Sequence `\0` is used as a string delimiter. The sequence `^P` induces the PIN printing.

As an example, consider sending the PIN to Mr Robert J Smith. It is advised to print both the customer's address as well as the PIN, in the following manner:

```
Robert J Smith
APT 4B                XXXX
39 ELM DR
MEDIA PA 19063
YOUR FULL SERVICE BANK
```

Where `XXXX` is the PIN and a final corporate message has been added. If there is similar information to be sent to another customers, that format (the position of the different printing fields) can form a template for all customers. Similarly, the corporate message is also the same for all types of printing. The easiest way to resolve all of this is including the corporate message in the template. (However, it is possible to include it as an extra printing field if preferred. This approach can be seen in the command 0301). Template will be in this case:

```
>L>013^0>L>013^1>041^P>L>013^2>L>013^3>L>013YOUR FULL SERVICE BANK>L>F
```

Where it can be seen how the sequence `^0` denotes the entry point for the first string in the data table, the sequence `^1` denotes the entry point for the second string in the table and so on. The sequence `^P` denotes the entry point for the PIN.

Therefore the data table to be included in this command will be:

```
Robert J Smith\0APT 4B\039 ELM DR\0MEDIA PA 19063
```

The \is used as a separator, and it is not necessary to include it after the last string. This string length is 00000049.

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The alternative is to position the message outside the template and include another unspecified string in the template (^4 in this chase). The string has to be sent to the printing command in that case for each print performed.

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3.10.2 1301 - Printing Template Loading

Load the template string for the subsequent printing commands.

Request Body:

Data	Length & Type	Description
Command Identifier	4H	**1301**.
Printing Template Length	3N	Next field length. Up to 400 characters.
Printing Template	varA	The printing template.

Response Body:

Data	Length & Type	Description
Command Identifier	4H	**1301**.
Error Code	8H	Error code received from the HSM: **00000000** - No error code. **xxxxxxxx** - Other code defined in Error Codes.

It is possible to distribute key components safely by printing them, for example, blind envelopes. To achieve that goal, you can connect Cryptosec Dekaton to the serial port of any printer.

Before using the print command, it is necessary to define the paper or envelope format. A format is maintained until it is overwritten by another. Two commands are used to send the format symbols to the module. The symbols that define either format fields or any fixed string are provided in the table presented below.

Command 1301 shall be used to create the document layout. The format strip can contain any fixed text, although it is suggested to restrict to formatting characters. This is because it is limited to 400 characters. The second command is integrated with the print function that makes use of the capabilities of printing. At that point you can include up to 16 text strings fixed to 252 characters each, involved with the printing according to the template set, together with the information supplied by the same command. Printing format symbols are included in the following table:

Symbol	Hex. Value	Description
>L	0x3E 0x4C	New line, carriage return
>V	0x3E 0x56	Vertical Tab
>H	0x3E 0x48	Horizontal Tab

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Symbol	Hex. Value	Description
>F	0x3E 0x46	Page break
>nnn	0x3E 0x3n 0x3n 0x3n	Goes to the column nnn from the left margin, where nnn is a 3 digit decimal number
^M	0x5E 0x49	Prints the third decrypted key component of a given key.
^P	0x5E 0x50	Prints decrypted PIN for envelope 1, or print a decrypted key component.
^Q	0x5E 0x51	Prints decrypted PIN for envelope 2, or print a decrypted key component.
^R	0x5E 0x52	Prints the reference for envelope 1
^S	0x5E 0x53	Prints the reference for envelope 2
^T	0x5E 0x54	Prints the last 6 digits of a bank account number for envelope 1, or prints the KCV of a key component
^U	0x5E 0x55	Prints the last 6 digits of a bank account number for envelope 2
|<L><hh…hh>	0x7C <L><hh…hh>	Send binary data to the printer, e.g. a control sequence. L contains the number of bytes to send, up to 255 bytes, followed by the data bytes to send.
^0	0x5E 0x30	Inserts print field 0 sent in the function that uses printing
^1	0x5E 0x31	Inserts print field 1 sent in the function that uses printing
..		
^F	0x5E 0x46	Inserts print field 15 sent in the function that uses printing

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3.11 Miscellaneous Commands

3.11.1 0904 - Securing Password

Encrypt a given password for secure storage externally.

Request Body:

Data	Length & Type	Description
Command Identifier	4H	**0904**.
Encryption Key	varK	The encryption key to protect the password (under LMK17).
Delimiter	1A	**H**. ***Only if the following password will be informed in hexadecimal.***
Password Length	2/4N	Next field length.[1]
Password	varB/H	The password to protect.

[1] **2** if the password is in binary format, **4** otherwise.

Response Body:

Data	Length & Type	Description
Command Identifier	4H	**0904**.
Error Code	8H	Error code received from the HSM: **00000000** - No error code. **xxxxxxxx** - Other code defined in Error Codes.
Protected Password Length	2/4N	Next field length.
Protected Password	varH	The protected password.

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3.11.2 0905 - Verify Password

Verify a password passing as many protected password as wanted.

Request Body:

Data	Length & Type	Description
Command Identifier	4H	**0905**.
Decryption Key	varK	The key to decrypt the protected password (under LMK17).
Delimiter	1A	**H**. ***Only if the following password will be informed in hexadecimal.***
Password Length	2/4N	Next field length.[1]
Password	varB/H	Password to be verified.
Number of Protected Passwords	2N	Number of protected passwords.
Protected Password Length	2/4N	Next field length.[2]
Protected Password	varH	The protected password.[2]

[1] **2** if the password is in binary format, **4** otherwise.
[2] The greyed fields will be repeated as many times as specified on field **Number of Passwords**.

Response Body:

Data	Length & Type	Description
Command Identifier	4H	**0905**.
Error Code	8H	Error code received from the HSM: **00000000** - No error code. **xxxxxxxx** - Other code defined in Error Codes.
Verification Result	1N	Verification result: **0** - Failed.[1] **1** - Correct.[2]

[1] None of the protected passwords has been verified successfully.
[2] One of the protected passwords has been verified successfully.

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3.11.3 1401 - HSM Test

Performs a test call to the HSM.

Request Body:

Data	Length & Type	Description
Command Identifier	4H	**1401**.

Response Body:

Data	Length & Type	Description
Command Identifier	4H	**1401**.
Error Code	8H	Error code received from the HSM: **00000000** - No error code. **xxxxxxxx** - Other code defined in Error Codes.
HSM Identfier	6H	First six digits from the HSM ID.

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3.11.4 1402 - HSM Certification Mode

Report the certification mode of the HSM.

Request Body:

Data	Length & Type	Description
Command Identifier	4H	**1402**.

Response Body:

Data	Length & Type	Description
Command Identifier	4H	**1402**.
Error Code	8H	Error code received from the HSM: **00000000** - No error code. **xxxxxxxx** - Other code defined in Error Codes.
Certification mode	1N	Certification mode configured: **0** - Non-Approved mode. **1** - PCI PTS HSM v4.0-Approved mode. **2** - NIST FIPS PUB 140-2-Approved mode.

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3.11.5 1403 - HSM API

Report the API configured on the HSM.

Request Body:

Data	Length & Type	Description
Command Identifier	4H	**1403**.

Response Body:

Data	Length & Type	Description
Command Identifier	4H	**1403**.
Error Code	8H	Error code received from the HSM: **00000000** - No error code. **xxxxxxxx** - Other code defined in Error Codes.
API	1N	API:[1] **0** - Legacy payment. **1** - LAN GP. **2** - Obsolete payment. **3** - PKCS#11 GP. **4** - Atalla payment. **5** - Advanced payment.

[1] Only values **0** or **1** are expected.

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3.11.6 1404 - HSM State

Report the state of the HSM.

Request Body:

Data	Length & Type	Description
Command Identifier	4H	**1404**.

Response Body:

Data	Length & Type	Description
Command Identifier	4H	**1404**.
Error Code	8H	Error code received from the HSM: **00000000** - No error code. **xxxxxxxx** - Other code defined in Error Codes.
State	1N	State: **1** - Not initialized. **2** - Initialized.

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3.11.7 1405 - HSM Console Session State

Report if a session is opened on the console of the HSM.

Request Body:

Data	Length & Type	Description
Command Identifier	4H	**1405**.

Response Body:

Data	Length & Type	Description
Command Identifier	4H	**1405**.
Error Code	8H	Error code received from the HSM: **00000000** - No error code. **xxxxxxxx** - Other code defined in Error Codes.
State	1N	Console session state: **0** - Production (Session not opened). **1** - Authorization (Session opened).

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3.11.8 1701 - Random Number Generation

Generates a random number.

Request Body:

Data	Length & Type	Description
Command Identifier	4H	**1701**.
Length	6N	Random number length in bits. Example: If 6 hexadecimal digits needs to be generate, '000048' should be that field's value (6 times 8 bits).

Response Body:

Data	Length & Type	Description
Command Identifier	4H	**1701**.
Error Code	8H	Error code received from the HSM: **00000000** - No error code. **xxxxxxxx** - Other code defined in Error Codes.
Random Number Length	6N	Next field length.
Random Number	varH	The generated random number.

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3.12 Key Diversification

3.12.1 1601 - Key Diversification

Derive a key according to one of the following algorithms:

Name	Algorithm
TDES-E-ECB	OutKey = InKey.encryptECB(DataField1 || DdataField2)
TDES-D-ECB	OutKey = InKey.decryptECB(DataField1 || DdataField2)
SIM	OutKey = Rightmost(InKey.encryptCBC(DataFieldIcc || DataFieldIcc || 0xFFFFFFFF, NULL IV))

Request Body:

Data	Length & Type	Description
Command Identifier	4H	**1601**.
Input Key	varK	Key to derive (under LMK: 2, 3, 6, 7, 8, 9, 10, 14 or 27).
Parity Check	1N	Check key parity: **0** - Do not check. **1** - Check odd parity. ***Only if the derivation key is under LMK9.***
Algorithm	2N	Derivation Algorithm: **12** - TDES-E-ECB. **15** - TDES-D-ECB. **22** - SIM Keys.
RFU	2N	Reserved for Future Use. ***Value: 00.***
Parity Adjustment	1N	Parity adjustment of diversified key: **0** - Do not adjust. **1** - Adjust odd parity. ***Only if the derivation key is under LMK9.***
Data Field 1 Length	2N	Next field length. ***Only if Algorithm is 12.***
Data Field 1	varH	The data field one. ***Only if Algorithm is 12.***
Data Field 2 Length	2N	Next field length. ***Only if Algorithm is 12.***
Data Field 2	varH	The data field two. ***Only if Algorithm is 12.***

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Data	Length & Type	Description
Icc Data	20H	Identification of the SIM (IccID). ***Only if Algorithm is 22.***
Internal Storage	1A	Optional. Diversified key internal storage: **S** - Store internally.

Response Body:

Data	Length & Type	Description
Command Identifier	4H	**1601**.
Error Code	8H	Error code received from the HSM: **00000000** - No error code. **xxxxxxxx** - Other code defined in Error Codes.
Output Key	varK	The derived key.

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3.12.2 1602 - Generate and Encrypt IPEK

Generates an IPEK key from a BDK.

The IPEK is to be loaded in a specific PIN pad and encrypted under the TK in variant method.

The diversification method used is DUKPT defined in [X9.24-1].

Request Body:

Data	Length & Type	Description
Command Identifier	4H	**1602**.
Base Derivation Key	varK	DES Base derivation key (under LMK19).
Transport Key	varK	Transport key (under LMK2).
KSN	20H	The Key Serial Number of the terminal.
Parity Adjustment	1N	Optional. IPEK parity adjustment:[1] **0** - Do not adjust. **1** - Adjust odd parity.

[1] If not present, the generated key may not have odd parity.

Response Body:

Data	Length & Type	Description
Command Identifier	4H	**1602**.
Error Code	8H	Error code received from the HSM: **00000000** - No error code. **xxxxxxxx** - Other code defined in Error Codes.
IPEK	varK	Generated Initial PIN Encryption key (under LMK28).

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3.12.3 1603 - Generate IPEK

Generates an IPEK key from a BDK in container v1 (Key Block). The IPEK can be exported in TR-31 format using the **0304** command.

The IPEK is to be loaded in a specific PIN pad and encrypted under the TK in variant method.

The diversification method used is DUKPT defined in [X9.24-1].

Request Body:

Data	Length & Type	Description
Command Identifier	4H	**1603**.
Base Derivation Key	varK	DES Base derivation key (under LMK19).
KSN	15H	The Key Serial Number of the terminal.[1]
Parity Adjustment	1N	Optional. IPEK parity adjustment:[2] **0** - Do not adjust. **1** - Adjust odd parity.

[1] Right justified and padded with 'Fs'. There is no Transaction Counter, as there would be with a KSN sent in a transaction data from a terminal.
Example: For a KSI+DID of 558812+99523313, the KSN field would be "F55881299523313".
[2] If not present, the generated key may not have odd parity.

Response Body:

Data	Length & Type	Description
Command Identifier	4H	**1603**.
Error Code	8H	Error code received from the HSM: **00000000** - No error code. **xxxxxxxx** - Other code defined in Error Codes.
IPEK	varK	Generated Initial PIN Encryption key (under LMK28).

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3.12.4 1604 - AES-DUKPT Derive Initial Key

Derive an AES initial key according to [X9.24-3, Derive Initial Key].

Request Body:

Data	Length & Type	Description
Command Identifier	4H	**1604**.
Base Derivation Key	varK	AES Base derivation key (under LMK19).
Initial Key ID	16H	Initial Key ID, concatenation of the BDK ID[1]and the Derivation ID[2].
LMK	2N	LMK index of the Initial Key. The value shall be **28**.[3]
Version	1N	Generated AES Initial Key container version: **0** - v0 container, unsupported. **1** - v1 container.

[1] The BDK ID is a 32-bit value that identifies the Base Derivation Key. This was formerly known as the Key Set ID (KSI) in the TDEA DUKPT specification.
[2] The Derivation ID is a 32-bit value that identifies the specific initial key derived under the Base Derivation Key specified by the BDK ID. This was formerly known as the 19-bit value Device ID (DID) or TRSM ID.
[3] To match AES-DUKPT Key Derivation Initial Key Key Usage.

Response Body:

Data	Length & Type	Description
Command Identifier	4H	**1604**.
Error Code	8H	Error code received from the HSM: **00000000** - No error code. **xxxxxxxx** - Other code defined in Error Codes.
Initial Key	varK	Generated AES Initial Key (under LMK28).

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3.12.5 1605 - AES-DUKPT Host Derive Working Key

Derive a symmetric working key according to [X9.24-3, Host Derive Working Key].

KSN compatibility mode:

[X9.24-3] defines a new format for the KSN, which is 16 bits longer than the legacy TDEA KSN. In cases where network infrastructure and existing software cannot easily accommodate the longer value, but designers would still like to take advantage of the increased cryptographic strength offered by the new version, it is defined an 80-bit KSN that is network-compatible. Note that it is NOT cryptographically compatible; it uses AES for key derivation, so it will not derive the same key(s) even if the same KSN is used.

For KSN compatibility mode, the host simply separates the 80 bit KSN into 56 bits of Initial Key ID and 21 bits of transaction counter. The Initial Key ID is padded to the right with zeros to create 64 bits. The transaction counter is padded to the left with zeros to create 32 bits.

Request Body:

Data	Length & Type	Description
Command Identifier	4H	**1605**.
Base Derivation Key	varK	AES Base derivation key (under LMK19).
Working Key Usage	2N	AES-DUKPT Key Usage for the working key: **00** - Key Encryption Key. **01** - PIN Encryption Key. **02** - Message Authentication, Generation Key.[1] **03** - Message Authentication, Verification Key.[1] **04** - Message Authentication, Both Ways Key. **05** - Data Encryption, Encrypt Key.[1] **06** - Data Encryption, Decrypt Key.[1] **07** - Data Encryption, Both Ways Key. **08** - Key Derivation Key. **09** - Key Derivation Initial Key.
Working Key Type	2N	AES-DUKPT Key Type for the working key: **00** - 2TDEA. **01** - 3TDEA. **02** - AES128. **03** - AES192.[2] **04** - AES256.[3] **05** - HMAC128. **06** - HMAC192.[2] **07** - HMAC256.[3]
Initial Key ID	16H	Initial Key ID, concatenation of the BDK ID[4] and the Derivation ID.[5]
Transaction Counter	8H	Counter that is increased for each transaction.

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Data	Length & Type	Description
LMK	2N	LMK index of the Working Key. Shall match Working Key Usage: **02** - Key Encryption Key. **03** - PIN Encryption Key. **08, 39, 41, 42 or 43** - Message Authentication, Generation Key. **08, 39, 41, 42 or 43** - Message Authentication, Verification Key. **08, 39, 41, 42 or 43** - Message Authentication, Both Ways Key. **07** - Data Encryption, Encrypt Key. **07** - Data Encryption, Decrypt Key. **07** - Data Encryption, Both Ways Key. **19** - Key Derivation Key. **28** - Key Derivation Initial Key.
Version	1N	Generated Symmetric Working Key version: **0** - v0 container, unsupported for AES working keys. **1** - v1 container.

[1] Not allowed in PCI PTS HSM v4.0-Approved mode.
[2] Not allowed when the Base Derivation Key is an AES128 key.
[3] Only allowed when the Base Derivation Key is an AES256 key.
[4] The BDK ID is a 32-bit value that identifies the Base Derivation Key. This was formerly known as the Key Set ID (KSI) in the TDEA DUKPT specification.
[5] The Derivation ID is a 32-bit value that identifies the specific initial key derived under the Base Derivation Key specified by the BDK ID. This was formerly known as the 19-bit value Device ID (DID) or TRSM ID in the TDEA DUKPT specification.

Response Body:

Data	Length & Type	Description
Command Identifier	4H	**1605**.
Error Code	8H	Error code received from the HSM: **00000000** - No error code. **xxxxxxxx** - Other code defined in Error Codes.
Working Key	varK	Generated Symmetric Working Key (under given LMK).

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3.12.6 1606 - ZKA Derivation

Derive a session key according to [ZKA, 1.9.3.2].

Request Body:

Data	Length & Type	Description
Command Identifier	4H	**1606**.
Base Derivation Key	varK	DES Base derivation key (under LMK19).
ZKA CV ID	1N	Fixed CV value identificator: **0** - MES (MAC). **1** - PAC (PIN encryption).
RND	16H	Random Number (16-bytes).
LMK	2N	LMK index of the Session Key. Shall match ZKA CV ID: **08** - MES (ISO 9797-1 MAC Algorithm 1). **08** - MES (ISO 9797-1 MAC Algorithm 3). **08** - MES (Any other MAC). **03** - PAC (PIN encryption).
Version	1N	Generated DES Session Key container version: **0** - v0 container. **1** - v1 container.

Response Body:

Data	Length & Type	Description
Command Identifier	4H	**1606**.
Error Code	8H	Error code received from the HSM: **00000000** - No error code. **xxxxxxxx** - Other code defined in Error Codes.
Session Key	varK	Generated DES ZKA Session Key (under given LMK).

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3.13 Digest

3.13.1 2000 - Hash Data Block

Compute a message digest function of large data blocks or devided in blocks.

Request Body:

Data	Length & Type	Description
Command Identifier	4H	**2000**.
Hash Identifier	2N	Hash algorithm: **01** - MD5. **10** - SHA-1. **11** - SHA-224. **12** - SHA-256. **13** - SHA-384. **14** - SHA-512
Block Index	2N	The hash step to perform: **01** - First block. **02** - Middle Block (status). **03** - Last block.
Data Format	2N	Format of the data to be informed: **01** - Hexadecimal. **02** - Binary.
Hash Status Length	6N	Next field length. ***Only if Block Index is 02 or 03.***
Hash Status	varH/B	Input state to continue or finish a previous hash process. ***Only if Block Index is 02 or 03.***
Message Length	6N	Next field length.
Message	varH/B	The message to digest.

Response Body:

Data	Length & Type	Description
Command Identifier	4H	**2000**.

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Data	Length & Type	Description
Error Code	8H	Error code received from the HSM: **00000000** - No error code. **xxxxxxxx** - Other code defined in Error Codes.
Data Format	2N	Format of the data to be informed: **01** - Hexadecimal. **02** - Binary. ***Only if Block Index is 01 or 02.***
Hash Status Length	6N	Next field length. ***Only if Block Index is 01 or 02.***
Hash Status	varH/B	The hash status. ***Only if Block Index is 01 or 02.***
Output Hash Length	6N	Next field length. ***Only if Block Index is 03.***
Output Hash	varH/B	The hash result. ***Only if Block Index is 03.***

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3.14 Decimalization Tables (Cryptosec-Banking Only)

3.14.1 2101 - Decimalization Table Update

Re-encrypt a Decimalization Table (DT) that is encrypted under a LMK from an old set. This makes it possible to upgrade all the DTs that are externally stored every the master key is changed.

For this command to work properly, the following items shall be taken into account:

- Have loaded new Master Keys into the HSM.
- Have kept the old LMKs of the previous Master Keys.
- The decimalization table to be updated shall be encypted under an old LMK: the update process is continuous because only the most recently replaced set of LMKs is stored.
- Not having deleted the old LMKs prior to the execution of this command. Once removed, it will only be possible to recover any decimalization table encrypted under this set of keys loading again the associated Master Keys.

Request Body:

Data	Length & Type	Description
Command Identifier	4H	**2101**.
DT	16/var[1] N/H/K[2]	Current Decimalization table.[3]

[1] Decimalization table length can be:
16 for a raw table.
var for a table in a container version 1.
[2] Decimalization table type can be:
N for a raw table in the clear.
H for an encrypted raw table.
K for a table in a container version 1, encrypted or in the clear.
[3] PCI PTS HSM v4.0-Approved mode of operation requires an encrypted raw table or a table in a container version 1. For new implementations, container version 1 is recommended.

Response Body:

Data	Length & Type	Description
Command Identifier	4H	**2101**.
Error Code	8H	Error code received from the HSM: **00000000** - No error code. **xxxxxxxx** - Other code defined in Error Codes.

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Data	Length & Type	Description
DT	16/var[1] N/H/K[2]	The decimalization table encrypted under the new LMK.[3]

[1] Decimalization table length can be:
16 for a raw table.
var for a table in a container version 1.

[2] Decimalization table type can be:
N for a raw table in the clear.
H for an encrypted raw table.
K for a table in a container version 1, encrypted or in the clear.

[3] PCI PTS HSM v4.0-Approved mode of operation requires an encrypted raw table or a table in a container version 1. For new implementations, container version 1 is recommended.

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3.15 X509 Certificates Management

3.15.1 2301 - X509 Request Generation

Generate a basic standard x509 certificate request (pkcs#10).

If none of the key labels exists, new RSA key pair will be generated and stored. Otherwise existing key will be used to generate the CSR.

Request Body:

Data	Length & Type	Description
Command Identifier	4H	**2301**.
Key Length	4N	Number of bits of the key to be generated. ***Only valid: 1024, 2048 and 4096.***
Public Exponent	1N	Public exponent of the key to be generated: **0** - 3. **1** - Fermat4.
Delimiter	1A	**;** (0x3B ASCII). ***Optional. If present, Hash Identifier shall be present. Otherwise SHA-256 will be used as signing hash algorithm.***
Hash Identifier	1N	Signing hash algorithm: **0** - MD5. **1** - SHA-1. **2** - SHA-224. **3** - SHA-256. **4** - SHA-384. **5** - SHA-512. ***Only present if Delimiter is present.***
Subject Attributes Count	2N	Number of attributes that made up the certificate's subject.
OID Length	2N	Next field length.[1] ***Value shall be informed in case of new or proprietary OIDs. If the OID to be used exists in the standard, the length of the field shall be zero (00) with no value at all.***
OID	varA	The attribute's OID.[1]
OID Short Name Length	2N	Next field length.[1] ***Value shall be informed in case of the OID exists in the standard. If the OID to be used is a proprietary OID, the length of the field shall be zero (00) with no value at all.***
OID Short Name	varA	The attribute's OID short name.[1]
Attribute Value Length	2N	Next field length.[1]

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Data	Length & Type	Description
Attribute Value	varA	The attribute's value.[1]
Password Challenge Length	2N	Next field length.
Password Challenge	varA	The password challenge.
Public Key Label Length	2N	Next field length.
Public Key Label	varA	The label under the public key is stored on the HSM.
Private Key Label Length	2N	Next field length.
Private Key Label	varA	The label under the private key is stored on the HSM.
Request Encoding Format	1N	The encoding format of the request is going to be generated: **0** - DER. **1** - PEM.

[1] The greyed fields will be repeated as many times as specified on field **Subject Attributes Count**.

Response Body:

Data	Length & Type	Description
Command Identifier	4H	**2301**.
Error Code	8H	Error code received from the HSM: **00000000** - No error code. **xxxxxxxx** - Other code defined in Error Codes.
Request Length	4N	Next field length.
Request	varB	The generated request.

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3.16 Legacy Commands

The commands in this category exist for legacy reasons. They have been superseded by newer (usually more generic) commands – refer to the individual commands for details. There are no plans to withdraw these commands, but it is strongly recommended that new application software use the newer commands.

3.16.1 0201 - Key Import

Import a key encrypted under a transport key.

For new implementations it is suggested the use of the 0202 command.
If the HSM is configured in PCI PTS HSM v4.0-Approved mode, the 0202 command SHALL be used.

Request Body:

Data	Length & Type	Description
Command Identifier	4H	**0201**.
Key Type	2N	Key type of the key to be imported.The value should be different from 00. ***Value according to Local Master Keys.***
Key To Import	varK	Key to import encrypted under the **Transport Key**.
Transport Key	varK	Decryption key to use (under LMK02, LMK60 or LMK62). ***LMK60 and LMK62 only allowed when the HSM is not configured in PCI PTS HSM v4.0-Approved mode.***

Response Body:

Data	Length & Type	Description
Command Identifier	4H	**0201**.
Error Code	8H	Error code received from the HSM: **00000000** - No error code. **xxxxxxxx** - Other code defined in Error Codes.
Imported key	varK	The secret/private key under the selected **Key Type**.

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3.16.2 0205 - Key Import in TR-31 Format

Import an external key which it is encrypted according to TR-31 standard into the HSM.

For new implementations it is suggested the use of the 0206 command.

Request Body:

Data	Length & Type	Description
Command Identifier	4H	**0205**.
Key Storage Type	1N	Key storage: **0** - External. **1** - Internal.
Key Container	1N	Key container version: **0** - v0 container. **1** - v1 container.
Transport Key	varK	Decrytion key (under LMK2 (not allowed in PCI PTS HSM v4.0-Approved mode) or LMK27).
Key	varT	Key to be imported encrypted according to TR-31 standard.[1]

[1] Please see [X9.TR-31, Annex A] and [TR-31 Implementation].

Response Body:

Data	Length & Type	Description
Command Identifier	4H	**0205**.
Error Code	8H	Error code received from the HSM: **00000000** - No error code. **xxxxxxxx** - Other code defined in Error Codes.
Imported Key	varK	The imported key.

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3.16.3 0302 - Key Export

Export a key encrypted with another key.

For new implementations it is suggested the use of the 0303 command.
If the HSM is configured in PCI PTS HSM v4.0-Approved mode, the 0303 command SHALL be used.

Request Body:

Data	Length & Type	Description
Command Identifier	4H	**0302**.
Key	varK	Key to export.
Transport Key	varK	Transport Key used to encrypt the key to export (under LMK 02, LMK 60 or LMK 62). ***LMK60 and LMK62 only allowed when the HSM is not configured in PCI PTS HSM v4.0-Approved mode.***

Response Body:

Data	Length & Type	Description
Command Identifier	4H	**0302**.
Error Code	8H	Error code received from the HSM: **00000000** - No error code. **xxxxxxxx** - Other code defined in Error Codes.
Exported Key	varK	The exported key encrypted under the **Transport Key**.

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3.16.4 0401 - Key KCV Calculation

For new implementations it is suggested the use of the 0404 command.

Computes the KCV of a symmetric key according to [X9.24-1]: The first six digits of the following operations are used as KCV:

- For DES keys the Legacy KCV computation method (encryption of a block of binary zeros) is used.
- For AES keys the CMAC KCV computation method (CMAC of a block of binary zeros) is used.

Request Body:

Data	Length & Type	Description
Command Identifier	4H	**0401**.
Key	varK	The key to get its check value.

Response Body:

Data	Length & Type	Description
Command Identifier	4H	**0401**.
Error Code	8H	Error code received from the HSM: **00000000** - No error code. **xxxxxxxx** - Other code defined in Error Codes.
KCV Length	2N	Next field length. **06**.
KCV	varH	Computed KCV.

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3.16.5 0504 - Script Signature (EMV 3.1.1) (Cryptosec-Banking Only)

Generate a script signature.

For new implementations it is suggested the use of the 0509 command.

Request Body:

Data	Length & Type	Description
Command Identifier	4H	**0504**.
Scheme	1N	The product: **0** - Visa. **1** - MasterCard.
SMI-MK	varK	SMI key (under LMK31 or LMK6).
PAN/PSN	16N	Pre-formatted PAN/PAN Sequence Number.
Integrity Session Data	16H	Data for integrity key session derivation.
Script Length	6N	Next field length.
Script	varH	Data to sign.

Response Body:

Data	Length & Type	Description
Command Identifier	4H	**0504**.
Error Code	8H	Error code received from the HSM: **00000000** - No error code. **xxxxxxxx** - Other code defined in Error Codes.
Signature	16H	The signature of the given script.

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3.16.6 0505 - Script Encryption (EMV 3.1.1) (Cryptosec-Banking Only)

Generate an encrypted script.

For new implementations it is suggested the use of the 0509 command.

Request Body:

Data	Length & Type	Description
Command Identifier	4H	**0505**.
Scheme	1N	The product: **0** - Visa. **1** - MasterCard.
SMC-MK	varK	SMC key (under LMK30 or LMK6).
PAN/PSN	16N	Pre-formatted PAN / PAN Sequence Number.
Integrity Session Data	16H	Data for confidentiality key session derivation.
Script Length	6N	Next field length.
Script	varH	Data to encrypt.

Response Body:

Data	Length & Type	Description
Command Identifier	4H	**0505**.
Error Code	8H	Error code received from the HSM: **00000000** - No error code. **xxxxxxxx** - Other code defined in Error Codes.
Encrypted Script Length	6N	Next field length.
Encrypted Script	varH	The encrypted script.

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3.16.7 0506 - PIN Exchange Script (EMV 3.1.1) (Cryptosec-Banking Only)

Generate a PIN exchange script.

For new implementations it is suggested the use of the 0509 command.

Request Body:

Data	Length & Type	Description
Command Identifier	4H	**0506**.
Scheme	1N	The product: **0** - Visa. **1** - MasterCard.
Transport PIN Block Key	varK	The source PIN encryption key, used to decrypt the supplied PIN block (under LMK3).
MK-AC Key	varK	MK-AC key (under LMK29 or LMK6). The Issuer Master Key to generate and verify application cryptograms. This is required to create PIN blocks for Visa PIN change. ***Only if Scheme is 0.***
MK-SMC	varK	The Master Key for Secure Messaging with Confidentially, used to encrypt the message (under LMK30 or LMK6).
MK-SMI	varK	The Master Key for Secure Messaging with Integrity, used to generate a MAC on the message (under LMK31 or LMK6).
Input PIN Block Format	1N	Input PIN block format: **1** - ISO1. **2** - ISO2. **4** - IBM3624. **5** - Diebold.
Output PIN Block Format	1N	Output PIN block format: **0** - PIN block I: standard EMV PIN block. **1** - PIN block II: VISA format without using current PIN. **2** - PIN block III: VISA format using current PIN.
New PIN Block	16H	New PIN block received from ATM.
Current PIN Block	16H	Current PIN block received from ATM. ***Only if Output PIN Block Format is 2.***
PAN/PSN	16N	Preformatted PAN and PAN Sequence Number. 14 most significant PAN digits on the rigth plus the PAN Sequence Number (PAN || PSN).

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Data	Length & Type	Description
Integrity Session Data	16H	Data for integrity session key derivation: ATC || 00 00 00 00 00 00 for Visa. RANDi Data for MasterCard.
Message Length	6N	Next field length.
Message	varH	Message to sign.

Response Body:

Data	Length & Type	Description
Command Identifier	4H	**0506**.
Error Code	8H	Error code received from the HSM: **00000000** - No error code. **xxxxxxxx** - Other code defined in Error Codes.
Re-encrypted PIN Block Length	6N	Next field length.
Re-encrypted PIN Block	varH	The re-encrypted PIN block data.
Script Signature	16H	MAC perform over the script command.

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3.16.8 0507 - PIN Exchange Script v2 (EMV 3.1.1) (Cryptosec-Banking Only)

Generate a PIN exchange script.

For new implementations it is suggested the use of the 0509 command.

Request Body:

Data	Length & Type	Description
Command Identifier	4H	**0507**.
Scheme	1N	The product: **0** - Visa. **1** - MasterCard.
Transport PIN Block Key	varK	The source PIN encryption key, used to decrypt the supplied PIN block (under LMK3).
MK-AC Key	varK	MK-AC key (under LMK29 or LMK6). The Issuer Master Key to generate and verify application cryptograms. This is required to create PIN blocks for Visa PIN change. ***Only if Scheme is 0.***
MK-SMC	varK	The Master Key for Secure Messaging with Confidentially, used to encrypt the message (under LMK30 or LMK6).
MK-SMI	varK	The Master Key for Secure Messaging with Integrity, used to generate a MAC on the message (under LMK31 or LMK6).
Input PIN Block Format	1N	Input PIN block format: **1** - ISO1. **2** - ISO2. **3** - ISO3. **4** - IBM3624. **5** - Diebold.
Output PIN Block Format	1N	Output PIN block format: **0** - PIN block I: standard EMV PIN block. **1** - PIN block II: VISA format without using current PIN. **2** - PIN block III: VISA format using current PIN. **3** - PIN block IV: Mastercard Pay Now & Pay Later.
New PIN Block	16H	New PIN block received from ATM.
Current PIN Block	16H	Current PIN block received from ATM. ***Only if Output PIN Block Format is 2.***

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Data	Length & Type	Description
PAN	16N	PAN. 16 right digits. If PAN length is less than 16, pad with zeros to the left.
PSN	2N	The PAN sequence number. **00..99**.
Integrity Session Data	16H	Data for integrity session key derivation: ATC || 00 00 00 00 00 00 for Visa. RANDi Data for MasterCard.
Message Length	6N	Next field length.
Message	varH	Message to sign.

Response Body:

Data	Length & Type	Description
Command Identifier	4H	**0507**.
Error Code	8H	Error code received from the HSM: **00000000** - No error code. **xxxxxxxx** - Other code defined in Error Codes.
Re-encrypted PIN Block Length	6N	Next field length.
Re-encrypted PIN Block	varH	The re-encrypted PIN block data.
Script Signature	16H	MAC performed over the script command.

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3.16.9 0601 - PIN Generation (Cryptosec-Banking Only)

Calculate a PIN according to a generation algorithm and return it as an encrypted PIN Block.

For new implementations it is suggested the use of the 0619 command.
If the HSM is configured in PCI PTS HSM v4.0-Approved mode, the 0619 command SHALL be used.

Request Body:

Data	Length & Type	Description
Command Identifier	4H	**0601**.
PIN Key	varK	PIN calculation key (under LMK45 (IBM German Bank Pool Institution or Interbank), LMK46 (IBM3624 or IBM3624 PIN Offset) or LMK4).
PIN Encryption Key	varK	PIN block transport key (under LMK3).
DT	16/var[1] N/H/K[2]	Decimalization table.[3]
PIN Block Format	1N	Output PIN block format: **0** - ISO0. **1** - ISO1. **2** - ISO2. **3** - ISO3. **4** - IBM3624. **5** - Diebold. **6** - Realsec1.
PIN Generation Algorithm	1N	PIN generation algorithm: **0** - IBM3624. **1** - IBM3624 PIN Offset. **4** - IBM German Bank Pool Institution. **5** - Interbank.
PAN Length	2N	Next field length.
PAN	varN	The PAN of the card.
Padding Value	1H	Padding value. ***Only if PIN Block Format is 4.***
PIN Length	1N	PIN length. ***Only if PIN Generation Algorithm is 0.***
Key Indicator	1H	Key indicator. ***Only if PIN Generation Algorithm is 5.***
Validation Field	3H	Validation field. ***Only if PIN Generation Algorithm is 5.***

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Data	Length & Type	Description
PIN Offset Length	1N	Next field length. ***Only if PIN Generation Algorithm is 1.***
PIN Offset	varH	The PIN offset. ***Only if PIN Generation Algorithm is 1.***

[1] Decimalization table length can be:
16 for a raw table.
var for a table in a container version 1.
[2] Decimalization table type can be:
N for a raw table in the clear.
H for an encrypted raw table.
K for a table in a container version 1, encrypted or in the clear.
[3] PCI PTS HSM v4.0-Approved mode of operation requires an encrypted raw table or a table in a container version 1. For new implementations, container version 1 is recommended.

Response Body:

Data	Length & Type	Description
Command Identifier	4H	**0601**.
Error Code	8H	Error code received from the HSM: **00000000** - No error code. **xxxxxxxx** - Other code defined in Error Codes.
Encrypted PIN Block	16H	The PIN block.

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3.16.10 0604 - PIN Offset Generation (Cryptosec-Banking Only)

Generate a PIN offset using the selected algorithm.

For new implementations it is suggested the use of the 0618 command.

Request Body:

Data	Length & Type	Description
Command Identifier	4H	**0604**.
PIN Key	varK	PIN calculation key (under LMK45 (Netherlands PIN 1, or IBM German Bank Pool Institution), LMK46 (IBM3624 PIN Offset) or LMK4).
PIN Encryption Key	varK	PIN block transport key (under LMK3).
DT	16/var[1] N/H/K[2]	Decimalization table.[3]
PIN Block Format	1N	Output PIN block format: **0** - ISO0. **1** - ISO1. **2** - ISO2. **3** - ISO3. **4** - IBM3624. **5** - Diebold. **6** - Realsec1.
Algorithm	1N	PIN generation algorithm: **1** - IBM3624 PIN Offset. **2** - Netherlands PIN 1. **4** - IBM German Bank Pool Institution.
PIN Block	16H	Encrypted PIN block.
PAN Length	2N	Next field length.
PAN	varN	The PAN of the card.
Padding Value	2H	The padding value use to create the PIN block. ***Only if PIN Block Format is 4.***
PIN Length	1N	Number of digits of the PIN. ***Only if Algorithm is 1.***

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Data	Length & Type	Description

[1] Decimalization table length can be:
16 for a raw table.
var for a table in a container version 1.

[2] Decimalization table type can be:
N for a raw table in the clear.
H for an encrypted raw table.
K for a table in a container version 1, encrypted or in the clear.

[3] PCI PTS HSM v4.0-Approved mode of operation requires an encrypted raw table or a table in a container version 1. For new implementations, container version 1 is recommended.

Response Body:

Data	Length & Type	Description
Command Identifier	4H	**0604**.
Error Code	8H	Error code received from the HSM: **00000000** - No error code. **xxxxxxxx** - Other code defined in Error Codes.
Offset	4N	Returned offset value.

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3.16.11 0614 - PIN Translation Using DUKPT (Cryptosec-Banking Only)

Translate a PIN encrypted under a derivation key to encryption under a transport key for transmission to another node.

The ANSI X9.24-1:2009 method for DUKPT key derivation can be used.

For new implementations it is suggested the use of the 0613 command.

Request Body:

Data	Length & Type	Description
Command Identifier	4H	**0614**.
PIN Key	varK	The PIN calculation key (under LMK4).
Base Derivation Key	varK	The Base derivation key (under LMK19).
Source PIN Block Format	1N	The source PIN block format: **0** - ISO0. **1** - ISO1. **2** - ISO2. **3** - ISO3. **4** - IBM3624. **5** - Diebold. **6** - Realsec1.
Destination PIN Block Format	1N	The destination PIN block format: **0** - ISO0. **1** - ISO1. **2** - ISO2. **3** - ISO3. **4** - IBM3624. **5** - Diebold. **6** - Realsec1.
Source PIN Block	16H	Encrypted PIN block.
Source PAN Length	2N	Next field length.
Source PAN	varN	The PAN of the **Source PIN Block**.
Random Length	2N	Next field length. ***Only if Destination PIN Block Format is 1 or 3.***
Random	varN	Random number used to form **Encrypted PIN Block**. ***Only if Destination PIN Block Format is 1 or 3.***
Padding	1N	Padding value. ***Only if Destination PIN Block Format is 4.***

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Data	Length & Type	Description
KSN	20H	Current Key Serial Number.

Response Body:

Data	Length & Type	Description
Command Identifier	4H	**0614**.
Error Code	8H	Error code received from the HSM: **00000000** - No error code. **xxxxxxxx** - Other code defined in Error Codes.
Encrypted PIN Block	16H	The PIN block.

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3.16.12 0702 - CVC3 Verification (Cryptosec-Banking Only)

Verify Mastercard CVC3 code.

For new implementations it is suggested the use of the 0704 command.

Request Body:

Data	Length & Type	Description
Command Identifier	4H	**0702**.
CVC Key	varK	CVV/CVC/CSS key (under LMK5).
Mandatory Legacy Field	1N	**1**.
PAN	16N	The personal account number.
UN	8H	Unpredictable Number.
ATC	4H	Application Transaction Counter.
Track Data Length	2N	Next field length.
Track Data	varH	Static Track (1 or 2) Data.
CVC3	3..5N	Last digits of the CVC3 to verify. From 3 to 5.

Response Body:

Data	Length & Type	Description
Command Identifier	4H	**0702**.
Error Code	8H	Error code received from the HSM: **00000000** - No error code. **xxxxxxxx** - Other code defined in Error Codes.
Validation Result	1N	Validation result: **0** Correct. **1** Failed.
Diagnostic Data	3N/5N	Computed CVC3. If Version is 0 this is 3N with the correct dCVV. If Version is 1, 2 or 3, this is 5N with the correct CVC3. ***Only if Validation failed and HSM is in Authorization mode.***

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3.16.13 0901 - Symmetric Encryption and Decryption

Symmetric (AES or 3DES) encryption and decryption.

It is responsibility of the host to send the precise amount of data. For encryption, no extra padding will be added to the message.

For new implementations it is suggested the use of the 0903 command.

Request Body:

Data	Length & Type	Description
Command Identifier	4H	**0901**.
Key	varK	The Encryption/Decryption key (under LMK7).
Direction	1N	Direction: **0** - Decrypt. **1** - Encrypt.
Mode	1N	Mode: **0** - CBC with zeroes initialization vector. **1** - ECB.
Message Length	6N	Next field length. ***It shall be multiple of 16 for 3DES keys or 32 for AES keys.***
Message	varH	The data to encrypt or decrypt.

Response Body:

Data	Length & Type	Description
Command Identifier	4H	**0901**.
Error Code	8H	Error code received from the HSM: **00000000** - No error code. **xxxxxxxx** - Other code defined in Error Codes.
Output Data Length	6N	Next field length.
Output Data	varH	Encrypted or decrypted data.

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3.16.14 0907 - Symmetric Encryption and Decryption TDES DUKPT

Symmetric encryption and decryption with DUKPT derivation.

It is responsible of the host to send the precise amount of data. For encryption, no extra padding will be added to the message.

For new implementations it is suggested the use of the 0911 command.

Request Body:

Data	Length & Type	Description
Command Identifier	4H	**0907**.
Direction	1N	Operation to perform: **0** - Encryption. **1** - Decryption.
Mode	1N	Encryption/Decryption mode to use: **0** - ECB. **1** - CBC with zeroes initialization vector.
Base Derivation Key	varK	Base derivation key (under LMK19).
KSN	20H	The Key Serial Number sent by the POS.
Message Length	6N	Next field length. ***Shall be multiple of 16.***
Message	varH	The message to encrypt or decrypt.

Response Body:

Data	Length & Type	Description
Command Identifier	4H	**0907**.
Error Code	8H	Error code received from the HSM: **00000000** - No error code. **xxxxxxxx** - Other code defined in Error Codes.
Output Data Length	6N	Next field length.
Output Data	varH	The message encrypted or decrypted.

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3.16.15 0909 - AES-GCM Encryption and Decryption

AES GCM encryption and decryption.

It is responsibility of the host to send the precise amount of data. For encryption, no extra padding will be added to the message.

For new implementations it is suggested the use of the 0912 command.

Request Body:

Data	Length & Type	Description
Command Identifier	4H	**0909**.
Direction	1N	Operation to perform: **0** - Decryption. **1** - Encryption.
Format	1N	Format of the data to be informed: **0** - Binary or Text. **1** - Hexadecimal.
Operational Key	varK	Encryption/Decryption key (under LMK7).
Initialization Vector	16B or 32H	Initialization vector.[1] ***Shall always exists, but it will only be used if the HSM is not configured in NIST FIPS PUB 140-2-Approved mode.***
Additional Authentication Data Length	6N	Next field length.[2]
Additional Authentication Data	varB/H	The Additional Authentication Data.[2]
Authentication Tag Length	2N	Next field length.[3] Valid Lengths: If **Format** is **0**: **04**, **08**, **12**, **13**, **14**, **15** or **16**. If **Format** is **1**: **08**, **16**, **24**, **26**, **28**, **30** or **32**.
Authentication Tag	varB/H	Authentication Tag. ***Only if Direction is 0.***
Message Length	6N	Next field length.[2]
Message	varB/H	The data to encrypt or decrypt.[2]

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Data	Length & Type	Description

[1] **16** if **Format** is **0** and **32** is **Format** is **1**.
[2] If **Format** is **0**, any value shall be valid. It shall be used as received.
If **Format** is **1**, it shall be multiple of 2. It shall be compressed before being used.
[3] If **Direction** is **0** Length of the Authentication Tag to be authenticated previous data decryption.
If **Direction** is **1** Length of the Authentication Tag to be returned along with the encrypted data.

Response Body:

Data	Length & Type	Description
Command Identifier	4H	**0909**.
Error Code	8H	Error code received from the HSM: **00000000** - No error code. **xxxxxxxx** - Other code defined in Error Codes.
Authentication Tag Length	2N	Next field length. ***Only if Direction is 1.***
Authentication Tag	varB/H	Authentication Tag.[1] ***Only if Direction is 1.***
Output Data Length	6N	Next field length.
Output Data	varB/H	Output data.

[1] Same length as specified at **Authentication Tag Length**.

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3.16.16 1001 - RSA-PKCS#1v1.5 Encryption and Decryption

RSA-PKCS#1v1.5 encryption and decryption.

For new implementations it is suggested the use of the 1003 command.

Request Body:

Data	Length & Type	Description
Command Identifier	4H	**1001**.
RSA Key	varK	The key to use (under LMK60 or LMK62). ***LMK62 only allowed when the HSM is not configured in PCI PTS HSM v4.0-Approved mode.***
Direction	1N	Operation to perform: **0** - Encrypt with public key. **1** - Decrypt with private key.
Message Length	6N	Next field length.
Message	varH	The data to encrypt or decrypt.

Response Body:

Data	Length & Type	Description
Command Identifier	4H	**1001**.
Error Code	8H	Error code received from the HSM: **00000000** - No error code. **xxxxxxxx** - Other code defined in Error Codes.
Output Data Length	6N	Next field length.
Output Data	varH	Encrypted or decrypted data.

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3.16.17 1002 - RSA Encryption and Decryption

RSA encryption and decryption.

In RSA-OAEP, SHA-1 is the hash algorithm used to calculate the digest of the encoding parameter and the hash algorithm used in the mask generation function.

For new implementations it is suggested the use of the 1003 command.

Request Body:

Data	Length & Type	Description
Command Identifier	4H	**1002**.
RSA Key	varK	The key to use (under LMK 60).
Direction	1N	Direction: **0** - Encrypt with public key. **1** - Decrypt with private key.
Algorithm	1N	Algorithm: **0** - RSA-PKCS#1 v1.5. **1** - RSA-OAEP.
Message Length	6N	Next field length.
Message	varH	The message to encrypt or decrypt.

Response Body:

Data	Length & Type	Description
Command Identifier	4H	**1002**.
Error Code	8H	Error code received from the HSM: **00000000** - No error code. **xxxxxxxx** - Other code defined in Error Codes.
Output data Length	6N	Next field length.
Output data	varH	The output data.

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3.16.18 1101 - RSA Signature and Verification

RSA signature and verification.

For new implementations it is suggested the use of the 1103 command.

Request Body:

Data	Length & Type	Description
Command Identifier	4H	**1101**.
RSA Key	varK	The key to use (under LMK61 or LMK62). ***LMK62 only allowed when the HSM is not configured in PCI PTS HSM v4.0-Approved mode.***
Direction	1N	Operation to perform: **0** - Sign with private key. **1** - Verify with public key.
Hash Identifier	1N	Hash algorithm to use: **0** - SHA-1. **1** - MD5. ***Only if Direction is 0.***
Algorithm	1N	Algorithm: **0** - RSA-PKCS#1 v1.5. **1** - X9.31. ***Only if Direction is 0.***
Message Length	6N	Next field length.
Message	varH	The message to sign or verify the given signature.
Signature Length	6N	Next field length. ***Only if Direction is 1.***
Signature	varH	The signature to be verified. ***Only if Direction is 1.***

Response Body:

Data	Length & Type	Description
Command Identifier	4H	**1101**.

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Data	Length & Type	Description
Error Code	8H	Error code received from the HSM: **00000000** - No error code. **xxxxxxxx** - Other code defined in Error Codes.
Signature Length	6N	Next field length. ***Only if Direction is 0.***
Signature	varH	The signature of the message. ***Only if Direction is 0.***
Verification Result	1N	The signature verification result: **0** - Correct. **1** - Failed. ***Only if Direction is 1.***

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Chapter 4

Error Codes

The following table defines the error codes and their respective definitions:

Table 4.1: Error Codes

Code	Definition
00000000	SUCCESSFUL EXECUTION.
00000001	MESSAGE FORMAT ERROR.
00000002	SERVICE UNAVAILABLE.
00000003	INVALID DATA.
00000004	IO ERROR.
00000005	ERROR UNSPECIFIC.
00000025	KEY NOT FOUND.
00001C00	MEMORY ERROR.
00006400	PAIR WISE CONSISTENCE TEST ERROR.
00006800	SIGN VERIFY TEST ERROR.
00007C00	WRONG NUMBER OF CUSTODIANS.
00009800	USER EXISTS.
0000A000	USER NOT EXISTS.
0000A800	INVALID LEN KEY.
0000B000	INVALID KEY.
0000B400	RSA KEYS GENERATION ERROR.
0000B800	INVALID DATA SIZE.
0000BC00	PUBLIC KEY CIPHER ERROR.
0000C000	WRONG ALGORITHM SELECTED.
0000CC00	KEY NOT EXISTS.
0000E000	WRONG COMMAND.
0000E500	DH ECIES ERROR.
0000F000	PASSWORD ERROR.
0000F400	MEMORY INTERNAL KEYS FULL.
0000FC00	KCV VERIFICATION ERROR.
00010000	WRONG PAD CHARACTER.

Error Codes (continued)

Code	Definition
00010400	WRONG PIN BLOCK.
00010800	INVALID LEN PIN.
00010C00	TOO MANY ONES.
00011000	INVALID LEN PAN.
00011400	INVALID LEN CVV.
00011800	WRONG PAN.
00011C00	WRONG PARAMETER.
00012000	WRONG PINBLOCK FORMAT.
00015C00	WRONG CONFIGURATION DATA.
00016000	WRONG CMM KEY.
00016400	WRONG OPTIONS INDICATOR.
00016800	WRONG METHOD INDICATOR.
00017000	VERIFICATION FAILED.
00017400	PRINT STRING TOO LONG.
00017C00	WRONG PKCS8 INFO.
00018000	WRONG DECIMALIZATION TABLE.
00018400	NOT ALLOWED IN PRODUCTION STATE.
00018800	INVALID LEN NA.
00019000	PRINTER NOT ENABLED.
00019400	OLD CMMS NOT EXIST.
00019800	PIN VERIFICATION LOCKED.
00019C00	PIN BLOCK LOCKED.
0001A400	CARD ACCESS ERROR.
0001A800	CARD WRONG TYPE.
0001AC00	CARD AUTHENTICATION ERROR.
0001B000	CARD PIN ERROR.
0001B400	CARD CAN NOT CHANGE PIN.
0001C400	CARD ALREADY UPDATED.
0001D800	USER ABSENT.
0001E100	LOG FILE NOT EXIST.
0001E200	LOG FILE READ ERROR.
0001E300	LOG FILE TOO BIG.
0001E400	UNMARSHALL ERROR.
0001E500	SIG FILE NOT EXIST.
0001E600	SIG FILE READ ERROR.
0001E700	SIG FILE TOO BIG.
0001EA00	LOG DB ACCESSFAIL.
0001EC00	IMPORT IN PROGRESS.
0001ED00	WRONG CONTAINER DESCRIPTION VERSION.
0001EE00	WRONG CONTAINER DESCRIPTION TYPE.
0001EF00	WRONG CONTAINER DESCRIPTION SIZE.
0001F000	WRONG CONTAINER DESCRIPTION MAC.
0001F100	WRONG CONTAINER DESCRIPTION PLAINTEXT.
0001F200	WRONG CONTAINER DESCRIPTION CIPHERTEXT.

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Error Codes (continued)

Code	Definition
0001F400	MAX SESSION NUMBER REACHED.
0001F500	USER DOES NOT EXIST.
0001F600	SESSION DOES NOT EXIST.
0001F700	USER ALREADY LOGGED IN.
0001F800	AUTHENTICATION FAIL.
0001F801	AUTHENTICATION REQUEST FAIL.
0001F802	AUTHENTICATION REQUEST NOT FOUND.
0001F803	AUTHENTICATION WRONG USER.
0001F900	USER NOT LOGGED IN.
0001FA00	SIGNATURE FAILED.
0001FB00	WRONG ATALLA AKB DELIMITERS.
0001FC00	WRONG ATALLA AKB HEADER.
0001FD00	WRONG ATALLA AKB ENCRYPTEDKEY.
0001FE00	WRONG ATALLA AKB MAC.
0001FF00	WRONG ATALLA AKB ENCRYPTEDKEY PAD.
00020000	SAME KEY ALREADY STORED.
00020100	WRONG CONTAINER DESCRIPTION TYPE.
000201A1	ERROR KEYBLOCK INCOMPATIBLE LMK SCHEMES.
000201A2	ERROR KEYBLOCK INCOMPATIBLE LMK IDS.
000201A3	ERROR KEYBLOCK INCOMPATIBLE KEYBLOCK LMK IDS.
000201A4	ERROR KEYBLOCK AUTHENTICATION FAILURE.
000201A5	ERROR KEYBLOCK INCOMPATIBLE KEY LENGTH.
000201A6	ERROR KEYBLOCK INVALID KEY USAGE.
000201A7	ERROR KEYBLOCK INVALID ALG.
000201A8	ERROR KEYBLOCK INVALID MODE OF USE.
000201A9	ERROR KEYBLOCK INVALID KEY VERSION NUMBER.
000201AA	ERROR KEYBLOCK INVALID EXPORTABILITY.
000201AB	ERROR KEYBLOCK INVALID NUMBER OPTIONAL BLOCKS.
000201D3	ERROR WRAPPING KEY LESS SECURE.
10100000	ERROR IN COMMAND FORMAT.
10100001	ERROR BAD MODE.
10100002	ERROR BAD TARGET VERSION.
10100003	ERROR DECODING OBJECT.
10100004	ERROR ENCODING OBJECT.
F00000C1	FILE OPEN ERROR.

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Appendix A

Acronyms

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Table A.1: Acronyms

Term	Definition
AAC	Application Authentication Cryptogram.
AAV	Accountholder Authentication Value.
AES	Advanced Encryption Standard.
ARPC	Authorization Response Cryptogram.
ARQC	Authorization Request Cryptogram.
ASCII	American Standard Code for Information Interchange.
ASN.1	Abstract Syntax Notation One.
ATC	Application Transaction Counter.
ATM	Automated Teller Machine.
ARC	Authorization Response Code.
BDK	Base Derivation Key.
CBC	Cipher Block Chaining.
CMAC	Cipher-based Message Authentication Code.
CMM	See **LMK**.
CPE	Customer-Provided Equipment.
CSC	Card Security Code.
CVC	Card Verification Code.
CVC3	Card Verification Code 3.
CVV	Card Verification Value.
DAC	Data Authentication Code.
DES	Data Encryption Standard.
DER	Distinguished Encoding Rules.
DT	Decimaliztion Table.
DUKPT	Derived Unique Key Per Transaction.
DH	Diffie Hellman.
DN	Dynamic Number.
ECB	Electronic Code Book.
EMV	Europay Mastercard Visa.

Acronyms (continued)

Term	Definition
FIPS PUB	Federal Information Processing Standards Publication.
GCM	Galois/Counter Mode.
HMAC	Keyed-hash Message Authentication Code.
HSM	Hardware Security Module.
IAV	Issuer Authentication Value.
IEC	International Electrotechnical Commission.
IO	Input Output.
IPEK	Initial PIN Encryption Key.
ISO	International Organization for Standardization.
iCVV	Integrated Chip Card Card Verification Value.
KCV	Key Check Value.
KSN	Key Serial Number.
LMK	Local Master Key.
MAC	Message Authentication Code.
MGF	Mask Generation Function.
MGF1	Mask Generation Function 1.
NIST	National Institute of Standards and Technology.
OAEP	Optimal Asymmetric Encryption Padding.
OID	Object Identifier.
PAN	Personal Account Number.
PCI	Payment Card Industry.
PEM	PIN Extraction Method.
PIN	Personal Identification Number.
PKCS	Public Key Cryptography Standards.
POS	Point Of Sale.
PSN	PAN Sequence Number.
PTS	PIN Transaction Security.
PVV	PIN Verification Value.
RSA	Rivest, Shamir and Adleman.
RND	Random Number.
SHA	Secure Hash Standard.
SIM	Subscriber Identity Module.
TC	Transaction Certificate.
TDEA	Triple Data Encryption Algorithm. See **TDES**.
TDES	Triple-DES.
TIBC	Tarjeta Inteligente de Bancos y Cajas.
TK	Terminal Key.
UKPT	Unique Key Per Transaction.
UN	Unpredictable Number.
ZKA	Zentraler Kredit Ausschuss.

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Appendix B

Contact information

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[X9.24-3] American National Standards Institute, *ANSI X9.24-3-2017.*

[ZKA] *Technischer Anhang zum Vertrag über die Zulassung als Netzbetreiber im electronic cash-System der deutschen Kreditwirtschaft, Version 5.3.1.*